HSPT Exam Simulator

Full-Length Practice Exam 10

Complete printable test with 298 multiple-choice questions, answer key, and detailed explanations.

Section	Questions	Time
Verbal Skills	60	20 minutes
Quantitative Skills	52	35 minutes
Reading Comprehension	62	35 minutes
Mathematics	64	45 minutes
Language	60	30 minutes

General Directions

Work through each section within the recommended time. Choose the best answer for every question. Mark one answer (A-D) for each item. Use the answer key and explanations only after completing the test.

Recommended total testing time: 165 minutes. Calculators should be used only when permitted by the rules followed for your HSPT preparation.

Section 1: Verbal Skills

60 questions | Recommended time: 20 minutes

1. Which word is closest in meaning to COPIOUS?

A) delicate
B) doubtful
C) abundant
D) distant

2. A RECEPTIVE listener is most likely to be:

A) eager to interrupt every speaker
B) unable to hear nearby sounds
C) determined to preserve every opinion unchanged
D) open to considering new ideas

3. An ABSTRUSE explanation is:

A) deliberately humorous
B) difficult to understand
C) brief but complete
D) based entirely on numbers

4. Which word is most nearly opposite in meaning to WISTFUL?

A) thoughtful
B) homesick
C) contented
D) longing

5. Which word is most nearly opposite in meaning to SCRUPULOUS?

A) meticulous
B) careless
C) principled
D) thorough

6. A material described as IMPERVIOUS to water is the opposite of a material that is:

A) resistant to water
B) permeable to water
C) unaffected by water
D) sealed against water

7. A screw thread changes turning motion into forward movement. Which device makes the most similar change?

A) A mirror that sends light back toward its source
B) A sleeve that protects a folded map
C) A rack-and-pinion mechanism that moves a straight toothed bar when its gear turns
D) A whistle that makes a sound when air passes through it

8. An anagram changes the order of letters while retaining every letter and its number of occurrences. Which action has the same relationship to its original material?

A) Shuffling a deck changes card order without adding or removing cards.

B) Sharpening a pencil removes some wood.

C) Mixing paints produces a new combined color.

D) Enlarging a photograph changes its dimensions.

9. A stencil can reproduce a letter without containing any ink. Which comparison has the same relationship between a guide and the material needed to make its product?

A) A candle contains the wax that it burns.

B) A cookie cutter defines a shape but supplies none of the dough.

C) A reservoir stores the water it releases.

D) A packet holds the seeds that are planted.

10. Which word shares the idea of TWO found in BILINGUAL, BIPEDAL, and BILATERAL?

A) biscuit

B) bison

C) bicolored

D) bitter

11. HEAD of a committee, HEART of a city, and FOOT of a mountain use body-part words figuratively. Which phrase belongs in that group?

A) the backbone of an argument

B) the palm of her hand

C) the neck of the singer

D) the elbow of the pitcher

12. Which group contains ONLY terms for a deliberately unfinished state?

A) draft, sketch, rough cut

B) verdict, settlement, final copy

C) origin, midpoint, destination

D) boundary, perimeter, circumference

13. A notice says: "This entrance is closed to deliveries, except flowers for the stage." A bouquet is being delivered for the stage. What does the notice establish?

A) Every other entrance must be locked.

B) All flowers anywhere may use this entrance.

C) The bouquet’s carrier must be an actor.

D) The bouquet is exempt from this delivery restriction.

14. A catalog lists a novel under its original title and again under its translated title. Both entries explicitly refer to the SAME work. What follows?

A) Two catalog entries need not represent two different works.

B) Every translated title has a different author.

C) The work was written twice.

D) A work can never have more than one title.

15. Every word on a card is either printed in black or printed in red. Every red word is underlined. At least one word is NOT underlined. What must be true?

A) Every black word lacks an underline.

B) There are more black words than red words.

C) No word is printed in red.

D) At least one word is printed in black.

16. To WIELD a tool is to:

A) hide it from view

B) hold and use it

C) repair its handle

D) exchange it for another

17. To WINNOW a long list of proposals is to:

A) memorize every word

B) put the oldest proposal first

C) remove less suitable choices

D) copy the list without changes

18. A GENTEEL manner is most nearly:

A) hurried and impatient

B) refined and politely behaved

C) secretive and suspicious

D) loud and demanding

19. ARABLE land is suitable for growing crops. Which description is most nearly opposite?

A) ready for planting

B) unfit for cultivation

C) capable of producing grain

D) available for farming

20. A CONTENTIOUS participant often starts arguments. Which description is most nearly opposite?

A) agreeable and cooperative

B) eager to dispute

C) quick to object

D) constantly quarrelsome

21. In describing a writing style, FLORID most nearly contrasts with:

A) ornate and elaborate

B) richly embellished

C) plain and restrained

D) highly decorative

22. Condensation changes water vapor into liquid. Which process completes the reverse transformation?

A) Freezing changes liquid water into ice.

B) Grinding breaks a solid into smaller pieces.

C) Filtering removes particles from a liquid.

D) Evaporation changes liquid water into vapor.

23. A confluence is a place where streams join into one flow. Which location has the most similar relationship to what meets there?

A) A summit above the surrounding valleys

B) A warehouse holding unopened packages

C) A balcony overlooking a courtyard

D) A junction where separate traffic routes merge

24. A pun can present the SAME expression with two possible meanings. Which visual device has the most similar structure?

A) An enlargement that makes every feature bigger

B) A border that covers a damaged edge

C) An ambiguous drawing that can be seen as either a vase or two faces

D) A copy that removes all identifying marks

25. Which word belongs with NEVER, SELDOM, and OFTEN because it describes how frequently something happens?

A) always

B) nearby

C) abroad

D) backward

26. RENT, LEASE, and LEND can describe temporary use without transfer of ownership. Which action instead transfers ownership to a purchaser?

A) hire

B) sell

C) borrow

D) rent

27. Which group contains ONLY verbs that can describe repeated back-and-forth movement?

A) rotate, ascend, expand

B) hover, rest, remain

C) depart, flee, escape

D) oscillate, shuttle, seesaw

28. A notice says visitors MAY request a guided tour. It gives no other rule about tours. What does MAY establish here?

A) Visitors have permission to request a tour.

B) Every visitor must request a tour.

C) Every requested tour must be free.

D) Tours are forbidden unless visitors arrive together.

29. A sealed envelope contains an accurate prediction, but nobody can verify WHEN the prediction was written. Which claim remains unproved?

A) The envelope contains writing.

B) The prediction agrees with the event.

C) The prediction was written before the predicted event.

D) The writing date has not been verified.

30. Every accepted form is either inspected or signed, but NOT both. Every signed form is sealed. An accepted form is NOT sealed. What must be true?

A) It was signed and inspected.

B) It was signed but not inspected.

C) It was neither inspected nor signed.

D) It was inspected and not signed.

31. A taunt intended to GOAD someone into replying is intended to:

A) reassure that person

B) inform that person

C) protect that person

D) provoke that person

32. An obstacle that STYMIES an effort most nearly:

A) blocks its progress

B) explains its purpose

C) celebrates its success

D) predicts its duration

33. A STOLID observer is most nearly:

A) restless and easily excited

B) impulsive and unpredictable

C) openly expressive and dramatic

D) calm and showing little emotion

34. After resting, her movements were no longer LANGUID. Which description is most nearly opposite to LANGUID?

A) sluggish

B) listless

C) vigorous

D) leisurely

35. Which description is most nearly opposite to TRITE, describing a remark?

A) stale and overused

B) fresh and original

C) familiar to the point of dullness

D) worn out through repetition

36. Which word is most nearly opposite to VORACIOUS, describing someone’s eating habits?

A) ravenous

B) gluttonous

C) abstemious

D) insatiable

37. A partition divides existing space into separate rooms without adding floor area. Which change is most similar?

A) A new page adds several new sentences.

B) A translation replaces one language with another.

C) A photocopy creates another complete document.

D) A paragraph break divides existing text without adding words.

38. A boat's wake can mark its route after the boat has moved ahead. Which example has the most similar relationship?

A) A contrail marks part of a plane's route after the plane has passed.

B) A weather vane shows the present wind direction.

C) A compass indicates north.

D) A beacon identifies its own fixed position.

39. For a tenant, rent is an expense; for the owner, the SAME payment is income. Which case similarly names a SINGLE TRANSFER differently at its two ends?

A) A box is empty at noon and full at dusk.

B) A faded sign is repainted in a brighter color.

C) A border crossing is a departure from one country and an arrival in the other.

D) Two audiences give different ratings to a concert.

40. Which word belongs with KNEAD, WRING, and CLENCH because each can describe an action using hand pressure?

A) listen

B) squeeze

C) glance

D) hum

41. Which relationship, like IS NEXT TO, necessarily works in both directions?

A) is the parent of

B) is older than

C) is a sibling of

D) is the supervisor of

42. Which group pairs each property with its denial, rather than merely with a different degree or an associated thing?

A) equal/unequal; complete/incomplete; occupied/unoccupied

B) warm/cold; big/small; early/late

C) painter/paint; singer/song; writer/ink

D) wheel/axle; window/curtain; page/book

43. A panel promises to CONSIDER every proposal. Later it rejects several after reviewing them. Which interpretation is consistent with the promise?

A) Rejecting any proposal proves that it was never reviewed.

B) Reviewing every proposal does not require approving every proposal.

C) The promise guarantees approval of every proposal.

D) Only proposals that are approved can be considered.

44. A codebook defines ZEL as NAR and NAR as ZEL, giving no examples or other definitions. Why might this fail to teach a new reader either meaning?

A) The two definitions prove that the terms are opposites.

B) Definitions must always appear in reverse alphabetical order.

C) A codebook cannot contain more than one unfamiliar term.

D) Each unfamiliar term is explained only through the other unfamiliar term.

45. Assume a statement must be either true or false, but not both. A card says only, "This statement is false." What follows when each truth assignment is checked?

A) Neither assignment consistently fits the statement’s own claim.

B) Calling it true makes the card’s claim consistent.

C) Calling it false confirms its claim is false.

D) Its truth depends only on the card’s color.

46. To DECRY a practice publicly is to:

A) describe it neutrally

B) condemn it openly

C) approve it enthusiastically

D) hide all mention of it

47. To ATTUNE a plan to local conditions is to:

A) announce it without explanation

B) repeat it without any change

C) adjust it to fit those conditions

D) reject every local suggestion

48. Her RUEFUL smile followed the realization that she had caused the delay. RUEFUL most nearly means:

A) triumphant

B) indifferent

C) threatening

D) regretful

49. Which description is most nearly opposite to RAUCOUS, describing a crowd’s voices?

A) harsh and noisy

B) loud and boisterous

C) rough and clamorous

D) quiet and subdued

50. The investigation exposed SORDID dealings marked by dishonesty. Which word is most nearly opposite to SORDID here?

A) honorable

B) shameful

C) degrading

D) disreputable

51. Which word is most nearly opposite to WILY, describing a person?

A) cunning

B) guileless

C) crafty

D) sly

52. A ruler's evenly spaced marks indicate equal intervals of LENGTH. Which example has the most similar relationship to TIME?

A) A steady metronome's beats mark equal time intervals.

B) An alarm announces one selected moment.

C) A clock's case protects its mechanism.

D) A calendar cover displays a picture.

53. A dimmer allows intermediate brightness settings, while a simple ON/OFF switch offers separate states. Which pair makes the corresponding contrast?

A) Two ramps have different slopes.

B) Two ladders have different numbers of rungs.

C) Two switches have differently colored buttons.

D) A ramp provides a continuous incline, while a staircase provides separate standing levels.

54. A room can keep the SAME number of people while some leave and others enter. Which example has the same underlying relationship?

A) A locked box retains its contents because nobody opens it.

B) A reservoir's water level stays constant because incoming water replaces outgoing water.

C) An empty bottle becomes full as water is added.

D) A picture changes color as more paint is applied.

55. Which word belongs with POCKET, DRAWER, and COMPARTMENT because it names a small storage space?

A) hinge

B) handle

C) cubby

D) latch

56. Which group contains ONLY words that can signal a contrast with, or concession to, a preceding statement?

A) because, therefore, hence

B) first, next, finally

C) however, nevertheless, nonetheless

D) here, nearby, elsewhere

57. Which group contains ONLY ways of joining material by adding a separate fastening material or object?

A) fold, twist, bend

B) carve, pare, whittle

C) sew, staple, glue

D) sort, arrange, align

58. One route planner selects the route with the FEWEST CHANGES. Another selects the one with the SHORTEST TRAVEL TIME. They recommend different routes. Why need this be no contradiction?

A) The fewest changes must always mean the shortest time.

B) Only one planner can define its selection rule.

C) Every route necessarily takes the same amount of time.

D) They judge routes by different criteria.

59. A forecast proves incorrect. Someone concludes that the forecaster must have deliberately lied. Which distinction is missing?

A) Being wrong does not by itself establish knowingly saying something false.

B) An incorrect prediction can never be compared with what happens.

C) Every prediction must be true when it is spoken.

D) A sincere prediction cannot turn out to be wrong.

60. A report concludes that ALL engineers prefer Tool Q after interviewing ONLY engineers who had already named Q as their favorite. Which criticism is strongest?

A) Every engineer excluded from the interviews must dislike Q.

B) The selection ensures favorable answers from those invited but does not establish everyone else’s preference.

C) The conclusion must be false merely because some engineers were not interviewed.

D) Engineers cannot have meaningful preferences about tools.

Section 2: Quantitative Skills

52 questions | Recommended time: 35 minutes

61. For each positive integer n, write the SUM of the digits of n^2. The list begins 1, 4, 9, 7, 7, 9, 13, 10. What is the ninth entry?

A) 18

B) 9

C) 17

D) 81

62. The numerators rise by 1. The denominators increase by 2, then 3, then 4, and so on. What comes next?
1/3, 2/5, 3/8, 4/12, ____

A) 5/17

B) 5/16

C) 5/18

D) 4/17

63. Starting with 37, alternately reverse the two digits and subtract 9. What comes next?
37, 73, 64, 46, 37, 73, ____

A) 37

B) 64

C) 46

D) 82

64. Replace each pair (a, b) with (a + b, a − b), using the OLD values for both calculations. What comes next?
(3, 1), (4, 2), (6, 2), (8, 4), (12, 4), ____

A) (16, 12)

B) (8, 16)

C) (16, 8)

D) (48, 3)

65. Insert parentheses into 2 × 3 + 4 × 5 without changing the order of numbers or operations. What is the GREATEST possible result?

A) 26

B) 46

C) 50

D) 70

66. What is the SMALLEST positive integer k for which 36/k is an odd integer?

A) 4

B) 2

C) 6

D) 12

67. Positive integers a and b satisfy a + b = 7. Which choice correctly describes the DIFFERENT values of a/b + b/a?

A) Six: every ordered pair gives a different value.

B) Three: 37/6, 29/10, and 25/12

C) One: the sum is always 2.

D) Two: only whether a < b or a > b matters.

68. Quantity A: The greatest possible remainder when a positive integer is divided by 8
Quantity B: The number of positive factors of 8

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

69. The number k is a nonzero real number.
Quantity A: $\sqrt{(k^2)}$
Quantity B: $-k$

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

70. Positive integers a and b leave remainders 5 and 6, respectively, when divided by 7.
Quantity A: The remainder when a × b is divided by 7
Quantity B: 2

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

71. Two adjacent angles form a straight angle. One is obtuse.
Quantity A: The measure of the other angle
Quantity B: The measure of the obtuse angle

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

72. Quantity A: The number of diagonals drawn from ONE vertex of a convex hexagon
Quantity B: The number of triangles in a pentagon triangulated by diagonals from ONE vertex

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

73. Two vertices at opposite ends of a rectangle's diagonal are (−2, −1) and (2, 1). Its sides need NOT be parallel to the coordinate axes.
Quantity A: The rectangle's perimeter
Quantity B: 12 units

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

74. Each new numeral inserts ONE additional zero between the same two 1s. What comes next?
11, 101, 1001, 10001, ____

A) 10001

B) 1000001

C) 100001

D) 110001

75. Entry n is the greatest common factor of n and 12. Entries 1 through 9 are shown. Which choice correctly gives entry 10 and its reason?
1, 2, 3, 4, 1, 6, 1, 4, 3, ____

A) 1: 10 and 12 share no positive factor greater than 1.

B) 2: it is the greatest factor shared by 10 and 12.

C) 4: it is the greatest power of 2 dividing 12.

D) 6: it is the largest proper factor of 12.

76. Begin with the list [0]. To double it, append a reversed copy of the OLD list with 1 added to every copied entry. The first twelve entries begin as shown. What fills the blank?
0, 1, 2, 1, 2, 3, 2, 1, 2, 3, 4, ____

A) 3

B) 0

C) 4

D) 5

77. For EVERY negative odd integer n, which expression gives the next SMALLER even integer?

A) n + 1

B) 2n

C) −n − 1

D) n − 1

78. Add every digit appearing in the ten integers 40 through 49. What is the total?

A) 45

B) 49

C) 85

D) 445

79. A four-digit number is written using the digits 1, 2, 3, and 4 exactly once. Without listing every arrangement, which statement must be true?

A) Every arrangement leaves remainder 1 when divided by 3.

B) Every arrangement is even.

C) Every arrangement is divisible by 3.

D) Every arrangement leaves remainder 2 when divided by 3.

80. In the five-digit number 4a7b2, a and b are digits and a = b + 3. The number is divisible by 9. What is the number?

A) 45722

B) 44712

C) 46732

D) 48752

81. Quantity A: The number of digits to the RIGHT of the decimal point in 0.0008
Quantity B: The number of NONZERO digits in 0.0008

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

82. Two positive numbers have mean less than 5.
Quantity A: The greater of the two numbers
Quantity B: 5

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

83. Expand (2x + 1)(x + 4) as a polynomial.
Quantity A: The sum of ALL its coefficients, including the constant term
Quantity B: 15

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

84. A convex quadrilateral has perimeter P and diagonal lengths d and e.
Quantity A: d + e
Quantity B: P

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

85. Quantity A: The number of rotational symmetries of a regular pentagon using turns GREATER than 0° and LESS than 360°
Quantity B: The number of reflection axes of a square

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

86. A rhombus has area 36 square units.
Quantity A: The SUM of its two diagonal lengths
Quantity B: 18 units

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

87. Start with 1. Multiply by 1/2, then by 2/3, then by 3/4, and continue that multiplier pattern. What comes next?
1, 1/2, 1/3, 1/4, ____

A) 1/8

B) 1/6

C) 2/5

D) 1/5

88. The successive increases are 1, 2, 6, 24, and then 120. Each increase is the preceding increase multiplied by the next integer. What comes next?
0, 1, 3, 9, 33, ____

A) 57

B) 120

C) 153

D) 165

89. Replace a number with the SUM of its DISTINCT prime factors, then add 1. Which choice gives the next term and the correct reason?
45, 9, 4, 3, 4, ____

A) 2: use the prime factor but omit the final addition.

B) 4: leave the current term unchanged.

C) 5: count the prime factor 2 twice, then add 1.

D) 3: count the distinct prime factor 2 once, then add 1.

90. Record an interval's midpoint. Starting with [0, 1], then keep its LEFT half, then the new interval's RIGHT half, alternating in that way. What is the next recorded midpoint?
1/2, 1/4, 3/8, 5/16, 11/32, ____

A) 21/64

B) 23/64

C) 3/8

D) 11/64

91. Which operation sends BOTH −3 and 3 to the SAME value?

A) Adding 3 to each number

B) Replacing each number with its square

C) Multiplying each number by 2

D) Changing each number's sign

92. Multiply all integers from −5 through 5 EXCEPT 0 and −2. What is the sign of the product?

A) negative

B) zero

C) positive

D) The sign cannot be determined.

93. A positive fraction in lowest terms is NOT an integer. Multiplying it by 6 gives an integer, and multiplying it by 15 also gives an integer. Which denominator can it have?

A) 2

B) 5

C) 30

D) 3

94. Two DISTINCT integers are marked on a number line.
Quantity A: The distance between their marks
Quantity B: 1/2

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

95. Two number cubes, each labeled 1 through 6, show an ODD total.
Quantity A: The larger shown value
Quantity B: 4

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

96. The decimal in Quantity A continues with 9s FOREVER.
Quantity A: 0.249999...
Quantity B: 1/4

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

97. A single straight cut crosses a convex paper polygon from one edge to another, without passing through any original vertex.
Quantity A: The total vertex count of BOTH pieces minus the original polygon's vertex count
Quantity B: 5

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

98. A regular hexagon and a square each have perimeter 12 units. An equilateral triangle with side s has area $(\sqrt{3}/4)s^2$.
Quantity A: The hexagon's area
Quantity B: The square's area

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

99. In triangle ABC, side BC is extended past C to D; B, C, D lie in that order.
Quantity A: Angle CAB
Quantity B: Angle ACD

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

100. List the positive factors of 24 in increasing order. Beside each factor, write 24 divided by that factor. What pair comes next?
(1, 24), (2, 12), (3, 8), (4, 6), ____

A) (6, 4)

B) (5, 5)

C) (6, 6)

D) (8, 3)

101. Entry n is the SMALLEST multiple of n that is AT LEAST 20. Entries 1 through 7 are shown. What is entry 8?
20, 20, 21, 20, 20, 24, 21, ____

A) 16

B) 20

C) 32

D) 24

102. Entry n counts all factors of 2 in the prime factorization of 1 × 2 × ... × n. Entries 1 through 7 are shown. What does entry 8 count?
0, 1, 1, 3, 3, 4, 4, ____

A) 5 factors of 2

B) 7 factors of 2

C) 6 factors of 2

D) 8 factors of 2

103. A machine records an input integer as (quotient, remainder) after division by 4. Which input is recorded as (3, 2)?

A) 14

B) 12

C) 10

D) 32

104. Two positive integers have a SUM equal to their PRODUCT. Which pair satisfies that condition?

A) 1 and 1

B) 2 and 3

C) 2 and 2

D) 3 and 3

105. Start with 3/4. In each step, DOUBLE the numerator and denominator, then subtract 1 from EACH of those doubled values. What fraction results after TWO steps?

A) 11/15

B) 5/7

C) 9/14

D) 9/13

106. Keep 1, 2, 3, 4, 5 in that order, with the first 1 positive. Insert either + or − in each of the four gaps; do not join digits. How many expressions have value 1?

A) Two expressions

B) No expressions

C) One expression

D) Four expressions

107. A finite data set contains at least one POSITIVE and one NEGATIVE number.
Quantity A: Its range
Quantity B: The largest absolute value of any number in the set

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

108. A three-digit palindrome reads the same forward and backward. Its digit sum is 12.
Quantity A: Its first digit
Quantity B: Its middle digit

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

109. The number n is a positive integer.
Quantity A: The number of positive ODD divisors of 2n
Quantity B: The number of positive ODD divisors of n

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

110. A triangle has THREE UNEQUAL side lengths.
Quantity A: Its number of reflection axes
Quantity B: 1

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

111. Five unit squares form one flat figure: one central square and one square sharing EACH of its four sides, with no overlapping interiors.
Quantity A: The figure's perimeter
Quantity B: 12 units

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

112. A rectangular prism's length is DOUBLED, leaving width and height unchanged. Then ALL THREE new dimensions are halved.
Quantity A: Its final volume
Quantity B: Its original volume

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

Section 3: Reading Comprehension

62 questions | Recommended time: 35 minutes

Literary Passage: The Other Handlebar

The tandem bicycle had two saddles and two sets of pedals. I chose the rear saddle because it looked higher. Only after Aunt Mina started down the lane did I discover that my handlebar did not steer.

"What is it for, then?" I asked.

"Holding on."

Hers decided every turn. Mine stayed stubbornly in place, a very convincing imitation of something important. I began to feel like an extra suitcase with knees.

"I seem to be ornamental," I announced.

Aunt Mina did not answer. Ahead of us, the lane climbed toward a stone bridge. Her shoulders bent forward, and the easy rhythm of our pedals slowed.

"A little more help back there?" she called.

I pushed harder. The bicycle moved more steadily, and we reached the bridge without stopping. At the top, she turned her head just far enough for me to see her smile.

"Thank you. I was beginning to think I would have to pedal for two."

On the way down, my handlebar still refused to steer. I tested it once, gently, then wrapped my hands around it and watched the road open beside her shoulder. At the next rise, I started pushing before she asked. This time I knew which part of the bicycle was listening to me.

113. What does Aunt Mina request as the bicycle approaches the bridge?

A) A different handlebar for her saddle

B) Directions to another lane

C) Permission to stop at a shop

D) More help with pedaling

114. Why does the narrator initially feel like extra luggage?

A) Aunt Mina has forbidden the narrator to pedal.

B) The fixed rear handlebar makes the narrator mistake lack of steering control for lack of usefulness.

C) The rear saddle has been removed.

D) The bicycle is carrying several actual suitcases.

115. When the narrator claims to be ornamental, the word means:

A) responsible for choosing the route

B) unable to remain seated

C) especially skilled at climbing hills

D) present for appearance rather than practical use

116. What is the effect of calling the narrator "an extra suitcase with knees"?

A) It establishes that the narrator is literally inside a suitcase.

B) It explains the bicycle’s exact weight limit.

C) It humorously conveys a feeling of being carried without contributing.

D) It proves that Mina cannot see the road.

117. How does the last paragraph develop the narrator's response to the fixed handlebar?

A) The narrator repairs the handlebar so it can steer.

B) The handlebar remains unchanged, but the narrator acts on a new understanding of the rear rider's role.

C) Mina gives up steering and takes the rear saddle.

D) The narrator decides to stop pedaling altogether.

Informational Passage: Faster Is Not More

A confectioner preparing syrup may crush sugar to help it dissolve quickly. Crushing exposes more surface to the water. Water can then reach more of the sugar at once, so small grains generally disappear faster than an equal mass in a large lump under otherwise similar conditions.

Speed, however, is not the same as the amount that can dissolve. For a given quantity of water at a fixed temperature, there is a limit to how much sugar can remain dissolved. Crushing does not by itself raise that limit. If the water has already reached it, adding powdered sugar will not make the extra sugar stay dissolved simply because its grains are small.

A solution at this limit is described as saturated. With excess sugar present, some solid remains after sufficient time, whether that excess began as powder or as large crystals. The solid might be harder to notice when its pieces are tiny, but invisibility at a glance is not proof that everything has dissolved.

The confectioner therefore asks two different questions: How soon must the syrup be ready, and how much sugar must it contain? Grain size can help answer the first. The second requires attention to the amount of water and its temperature. Confusing the questions can produce a faster process without the intended result.

118. According to the passage, why can crushing sugar speed its dissolving?

A) It exposes more sugar surface to the water.

B) It increases the quantity of water.

C) It automatically raises the water's temperature.

D) It changes sugar into a different substance.

119. What conditions does the passage hold fixed when discussing the dissolving limit?

A) The quantity of water and its temperature

B) The color of the container and the stirring spoon

C) The confectioner's age and experience

D) The original shape of every sugar crystal

120. Which statement best expresses the passage's central distinction?

A) Crushed sugar always remains solid.

B) Every syrup requires the largest available sugar crystals.

C) Dissolving speed alone determines final sugar concentration.

D) A change can speed dissolving without increasing how much can remain dissolved.

121. As used in the passage, saturated describes a solution that:

A) contains no water at all

B) must contain only large crystals

C) has reached its dissolving limit under the stated conditions

D) has just begun dissolving its first grain

122. Two syrups use equal water quantities at the same temperature, with excess sugar and sufficient time. Only the initial grain size differs. Which conclusion is supported?

A) The powdered sugar must all remain dissolved.

B) Both syrups must finish dissolving at exactly the same instant.

C) The finer starting grains alone do not justify a higher final dissolved amount.

D) The syrup with large crystals must contain no dissolved sugar.

Humanities Passage: Where a Monument Places the Visitor

Imagine two proposed monuments remembering the same voyage. Each design uses the same kind and quantity of stone. In one, a tall figure stands on a high pedestal, looking toward the horizon. In the other, the travelers' names run along a low wall, where visitors can stand close enough to trace individual letters.

The first design places the visitor below a commanding image. The upward gaze and distant horizon can emphasize determination, leadership, or an achievement viewed as a single dramatic act. The second distributes attention across many names. Visitors may move slowly along the wall, looking for a relative or noticing a name they have never heard.

These differences are not merely matters of decoration. Height, distance, and arrangement invite different kinds of attention. The stone is a kind of script for the visitor's movement, though nobody is forced to follow it exactly. Nor does either design dictate one inevitable emotion. A visitor might admire the tall figure, question whom it represents, or simply study the carving. Someone at the wall might feel connection, curiosity, or uncertainty. Design can guide an encounter without controlling every response.

Comparing the proposals therefore requires more than asking which uses finer stone. We should ask what each makes prominent, whose presence it records, and how it positions the people who come to remember.

123. Which feature belongs to the second proposed monument?

A) Names placed along a low wall within close reach

B) One figure standing on a high pedestal

C) A moving sculpture above a doorway

D) A wall that contains no individual names

124. What chiefly motivates the author’s comparison of the two monuments?

A) To prove that high monuments should never be built

B) To identify the actual leader of a historical voyage

C) To explain how monument design guides visitors’ attention and participation

D) To calculate the cost of carving different stones

125. Why does the author specify the same kind and quantity of stone for both proposals?

A) It proves that both monuments must be equally tall.

B) It directs comparison toward arrangement and experience rather than material differences.

C) It establishes that every visitor will prefer the wall.

D) It shows that neither monument can contain any carving.

126. In the last paragraph, prominent most nearly means:

A) physically hidden

B) especially noticeable or important

C) recently repaired

D) difficult to pronounce

127. What does calling the stone "a kind of script" suggest?

A) The stone must display the complete text of a play.

B) Every visitor is required to say the same words.

C) A monument can communicate only through carved sentences.

D) Its arrangement proposes movements and ways of attending, much as a script guides action.

128. Why does the author follow the design comparison with examples of varied visitor reactions?

A) To qualify the claim that design guides attention by showing that it does not control every response

B) To replace the comparison with directions for carving stone

C) To prove that the two designs are physically identical

D) To show that names cannot interest visitors

Literary Passage: A Suitable Occasion

Uncle Davin gave me an umbrella with tiny foxes painted inside its canopy. "Something better than clouds to look at," he said.

I decided it deserved a special occasion. A walk to the vegetable market did not qualify, but I brought it along because leaving it at home seemed ungrateful.

Halfway there, the rain began. I tucked the folded umbrella under my coat. The canopy remained pristine; I was not faring as well. Water ran from my hair into my collar, and the shopping list softened in my pocket.

At the market, the woman selling potatoes looked at my dripping sleeves.

"Doesn't it open?" she asked.

"Of course. It was a present."

"Ah," she said. "I thought it was an umbrella."

I examined the neat dry folds. Uncle Davin had spent an evening painting foxes that I had seen only once, when he first opened the canopy. Since then, I had protected them so successfully that they might as well have stayed in a cupboard.

I stepped away from the stall and pressed the catch. Above me, the foxes appeared among green leaves. Drops pattered on the other side, where they belonged.

The woman handed me the potatoes. "A suitable occasion?"

"An excellent one," I said, and walked home beneath my present.

129. What has Uncle Davin painted on the umbrella?

A) Potatoes on the handle

B) Clouds on the outside

C) A shopping list on one panel

D) Tiny foxes inside the canopy

130. Why does the stallkeeper reply, "I thought it was an umbrella"?

A) She believes the object is actually a walking stick.

B) She points out that the narrator’s treatment of the gift defeats its ordinary purpose.

C) She cannot recognize any folded umbrella.

D) She is asking who painted the foxes.

131. In "I was not faring as well," faring refers to how the narrator is:

A) doing under the circumstances

B) paying for travel

C) folding the umbrella

D) judging the potatoes

132. What creates the central situational irony?

A) The potatoes are sold at a market.

B) The narrator gets soaked while keeping an umbrella dry.

C) The shopping list contains written words.

D) The umbrella has a catch that opens it.

133. How does the ending change the meaning of a "special occasion" for the narrator?

A) Only a formal celebration can justify opening the umbrella.

B) The gift becomes valuable only after its paintings disappear.

C) A useful ordinary moment becomes a worthy occasion for enjoying the gift.

D) The narrator decides that presents should remain unopened.

Informational Passage: Gas Inside the Dough

A ball of bread dough can grow much larger while it rests, even though nobody adds another scoop of flour. The change is not simply a collection of yeast cells growing into a solid mass the size of a loaf. Much of the increased volume comes from gas held inside the dough.

Yeast uses sugars and releases carbon dioxide. Small gas pockets can expand within a stretchy network formed by proteins in the dough. A useful model is a crowd of tiny balloons: gas occupies space, while the surrounding material must stretch enough to accommodate it and remain intact enough to hold it.

Producing gas and retaining gas are therefore different requirements. If gas escapes readily, active yeast alone cannot guarantee a large increase in volume. Conversely, a dough's appearance does not reveal exactly how many yeast cells it contains. Its ability to retain gas also affects the result.

During baking, the pockets can expand further, and the dough's structure eventually becomes firmer. Other changes occur too, so yeast is not a complete explanation of everything that happens in an oven. Still, the distinction between gas production and gas retention explains why a rising dough is more than an expanding pile of ingredients. Its changing structure matters along with the activity inside it.

134. Which gas does the passage identify as a product of yeast using sugars?

A) Carbon dioxide

B) Oxygen

C) Helium

D) Nitrogen

135. Which explanation best captures the passage’s main idea about rising dough?

A) The flour grains themselves must become larger.

B) Yeast cells form a solid block with no gas spaces.

C) Gas production and a structure capable of retaining that gas both contribute.

D) Any amount of active yeast guarantees the same loaf volume.

136. Two doughs have equal volumes after resting. Why can the passage NOT establish that they contain equal numbers of yeast cells?

A) Yeast never contributes to rising.

B) Equal volume proves both doughs contain no gas.

C) The passage says proteins cannot stretch.

D) Differences in gas retention can affect volume as well as yeast activity.

137. How does the explanation mainly develop?

A) It ranks bread types by price and then describes their colors.

B) It presents a complete recipe followed by a shopping list.

C) It introduces the volume change, explains gas production and retention, and briefly follows the structure into baking.

D) It describes only the outer crust and never discusses internal spaces.

138. What does the comparison with tiny balloons help readers understand?

A) Gas needs enclosing material that can stretch while still containing it.

B) Each yeast cell is literally a manufactured balloon.

C) The loaf's ingredients must be replaced by rubber.

D) Gas can remain trapped without any surrounding structure.

Informational Passage: Water in Narrow Spaces

Place the lower end of a clean, narrow glass tube in water, and water can rise inside it above the level outside. No pump is hidden in the tube. The effect involves interactions among the water, the glass, and gravity.

Water is attracted to the glass surface, an interaction called adhesion. Water molecules also attract one another; this is cohesion. Near the glass, these interactions help produce a curved water surface. The forces at that surface can support a small column of water against its weight.

The rising column becomes heavier as it grows taller. Eventually, its weight balances the upward supporting effect, and the rise stops. Thus, the observation does not show that gravity has disappeared or that water can climb to any height.

Under comparable conditions with water and clean glass, a narrower tube supports a higher rise than a wider one. The narrower column has less water to support relative to the amount of contact around its edge.

Similar effects help water spread through the narrow spaces in a paper towel. Those spaces form passages among fibers; the water does not need to flow through the solid material of each fiber. When spreading occurs horizontally, it need not lift an increasingly tall column against gravity. One mechanism can therefore appear in different arrangements, whose limits depend partly on their geometry.

139. Which term names the attraction between water and the glass surface?

A) Cohesion

B) Gravity

C) Evaporation

D) Adhesion

140. According to the passage, what eventually stops the upward rise in the tube?

A) Gravity permanently stops acting.

B) The water column's weight balances the upward supporting effect.

C) The glass changes into a solid barrier across the opening.

D) Every water molecule loses its attraction to glass.

141. Why does the author compare narrow and wide clean glass tubes?

A) To claim that wide tubes contain no glass

B) To show that geometry changes how much water must be supported relative to the contact around the edge

C) To establish that every liquid behaves identically

D) To explain why water is heavier than glass

142. Water spreads along a horizontal towel farther than it rises vertically in one arrangement. Which interpretation is best supported?

A) Horizontal spreading proves that cohesion has vanished.

B) Water must be flowing through solid glass in both cases.

C) The same type of interaction can operate while different geometric arrangements impose different limits.

D) Any difference proves that one observation is impossible.

143. In the final paragraph, passages most nearly means:

A) routes or channels through which water can move

B) sections of a written book

C) permissions granted to visitors

D) completed periods of time

144. Which statement best summarizes the author’s explanation of water rising without a pump?

A) Water in a narrow space becomes weightless.

B) A paper towel must secretly contain a mechanical pump.

C) Water always rises to the same height regardless of its surroundings.

D) Surface interactions can support water, while weight and geometry limit the result.

Literary Passage: For the Next Attendant

To the next carousel attendant:
The ticket punch sticks if you feed a card into it at an angle. Keep the corners straight, and you will have no trouble. There are spare rolls in the lower drawer, underneath the cloth with the yellow stripe.
That is the practical part. You will find the rest in the manager's manual.
There are a few trifles it leaves out. A little boy in a green cap calls the blue horse a whale. He knows perfectly well it is a horse. He explained that a creature traveling around the world all afternoon ought to have a bigger name.
Mr. Bexley counts his change twice. His grandson likes helping, so let them finish before serving the next person. They are not doubting your arithmetic.
The stone on the counter is there to keep the ticket rolls from wandering when the sea wind gets through the window. It came from the beach beyond the red steps. I had intended to take it home, but you may find it useful.
The blue cushion belongs to the chair now. I shall not need it.
If another pair of hands would help on your first Saturday, I could probably come by. Not because I shall miss the place, of course. The ticket punch can be difficult until you get used to it.
Best wishes,
Odette

145. Where does the note say the spare ticket rolls are kept?

A) Under the blue cushion

B) Beyond the red steps

C) Inside the manager’s coat

D) In the lower drawer beneath a yellow-striped cloth

146. What do Odette’s observations about the visitors most strongly suggest?

A) She has paid personal attention to the people who return to the carousel.

B) She has never met the visitors.

C) She believes every customer doubts the arithmetic.

D) She wants the replacement to refuse unusual requests.

147. In "a few trifles it leaves out," trifles most nearly means:

A) expensive prizes

B) serious accusations

C) written contracts

D) small matters

148. What creates the gentle irony in Odette’s claim that she will not miss the place?

A) Her detailed memories and offer to return suggest attachment despite her denial.

B) She admits that the carousel has never operated.

C) She proves the horse is actually a whale.

D) She refuses to leave any useful information.

149. How does the note’s movement from equipment instructions to personal observations affect its meaning?

A) It shows that the writer cannot locate the equipment.

B) It turns every personal observation into an official rule.

C) It gradually reveals that handing over the job also means leaving familiar relationships and routines.

D) It abandons the handover to describe an unrelated journey.

Literary Passage: The First New Mark

Before the arena doors opened, Lenka drove the resurfacing machine onto the ice. The empty seats rose around her in gray banks. Without an audience, the small clatter of a loose latch traveled farther than applause seemed to travel the night before.

She began at the boards. Behind the machine, a glossy strip replaced the scratched white surface. Thin curls of scraped ice gathered inside the machine, while a fresh film of water spread behind it. The overhead lights stretched into long pale bars across the wet track.

On each circuit, the strip widened. A deep turning mark vanished near the center line. Another pass erased the fan of scratches where a player had stopped hard beside the goal. Last night's swift decisions were losing their outlines.

Lenka watched the next turn rather than the disappearing marks. The motor kept its low, even note. Beyond the glass, a cleaner moved through the aisle, lifting paper cups from beneath the seats.

After the surface had set and the doors opened, the first skater stepped out alone. For a moment, the glossy expanse held the lights without interruption. Then one blade drew a curve through their reflections.

Lenka paused beside the open equipment gate. From where she stood, that first white curve looked like a question mark. The rest of the morning had not yet answered it.

150. What happens to the deep turning mark near the center line?

A) Lenka paints a circle around it.

B) It grows deeper after every pass.

C) The first skater creates it before the doors open.

D) One pass of the machine removes its visible outline.

151. Which description best captures the passage’s main focus?

A) An argument over last night’s score

B) The quiet preparation of a used arena for the next period of activity

C) Instructions for choosing a winning hockey team

D) A breakdown that prevents the arena from opening

152. If a visitor saw only the smooth surface just before the first skater entered, which conclusion would the passage warn against?

A) The overhead lights were on.

B) The ice had never been used before that morning.

C) The surface had been prepared.

D) The doors had opened after Lenka’s work.

153. What does the final image of a question mark chiefly suggest?

A) The skate blade has written an actual request for instructions.

B) Lenka has forgotten the arena’s location.

C) The surface is too damaged for anyone to skate.

D) The next activity is beginning, while its particular course remains unwritten.

154. Why does the narrative include erased traces of last night before showing the first new curve?

A) It proves that the new skater caused all the earlier scratches.

B) It places the first skater before Lenka’s arrival.

C) The sequence makes the renewed surface a transition between completed activity and an unknown next chapter.

D) It explains why the cleaner must stop collecting cups.

Informational Passage: A Bell That Interrupts Itself

In one simple electric bell, holding down a button produces a rapid series of strikes. Yet current does not flow continuously through the bell's electromagnet. A moving contact repeatedly interrupts it.

When the button is held and the contact is closed, current flows through a coil. The coil acts as an electromagnet and pulls a spring-mounted metal arm toward it. A striker attached to the arm hits the bell. That same movement separates the contact surfaces, opening the circuit.

The broken circuit stops current through the coil, so the magnetic pull disappears. The spring then returns the arm toward its starting position. This closes the contact again. With the button still held, current can resume, the electromagnet pulls again, and the cycle repeats.

The interruption is therefore part of normal operation, rather than evidence that the bell has failed. If the arm were held against the electromagnet and the circuit stayed closed, the repeated release-and-return motion would not occur in the described way.

Nor does the bell create its own supply of energy. An electrical source provides energy during the connected portions of the cycle. The spring helps return the arm, but it does not replace that source. The design makes one sustained request at the button produce many brief electrical and mechanical events inside the bell.

155. In this bell, what movement opens the circuit?

A) The button is released after every strike.

B) The moving arm separates the contact surfaces.

C) The spring removes the coil from the bell.

D) The electrical source changes into a magnet.

156. What chiefly explains the repeated strikes while the button stays pressed?

A) The moving arm repeatedly breaks and restores the coil’s circuit.

B) An unchanging magnetic pull keeps the striker permanently in motion.

C) The spring manufactures electrical energy.

D) Someone must press and release the button for each strike.

157. Suppose the contact surfaces could not separate when the arm moved. What would be best supported by this description?

A) The described break-and-return cycle would be prevented.

B) The bell would no longer require an energy source.

C) The spring would necessarily become an electromagnet.

D) Each button press would automatically produce a louder strike.

158. In the phrase "a moving contact," contact refers to:

A) a person who supplies information

B) a written agreement

C) a conducting part that touches another part to complete a circuit

D) an appointment for a conversation

159. Why does the author contrast the sustained button press with the brief events inside the bell?

A) To show that the button must be pressed once per strike

B) To distinguish one continuing external request from the repeated internal switching it produces

C) To deny that any part of the bell moves

D) To establish that the spring supplies all required energy

Literary Passage: Priority

[At a ferry terminal. Vesa approaches a counter holding a ticket. Beyond it, a long line curves toward the boarding gate.]
VESA: Does this priority ticket let me board before the other passengers?
CLERK: Before passengers with ordinary tickets, certainly.
VESA: Then I paid extra for a privilege.
CLERK: Exactly.
[Vesa joins the line. She studies the tickets ahead of her, then returns to the counter.]
VESA: Everyone in that line has the same gold stripe.
CLERK: The priority stripe. A very popular choice today.
VESA: Where are the ordinary passengers?
CLERK: Apparently, there aren't any on this sailing.
VESA: So whom do I board before?
CLERK: Anyone with an ordinary ticket.
VESA: But there isn't anyone.
CLERK: Your privilege remains exclusive to priority holders.
[Vesa looks from her ticket to the line.]
VESA: And every passenger is a priority holder.
CLERK: There is a priority-plus option.
VESA: Does that put me ahead of priority tickets?
CLERK: It does.
VESA: Can everybody buy that too?
CLERK: We would not wish to disappoint anyone.
[Vesa folds her ticket.]
VESA: I think I have bought enough ways to remain where I am.
CLERK: You don't want another upgrade?
VESA: Not if I have to bring the whole line with me.
[She returns to the place she left. The line has not moved.]

160. What does Vesa discover about the long boarding line?

A) It consists entirely of passengers without tickets.

B) It leads to a different ferry.

C) Every passenger in it has a priority ticket.

D) Only the clerk may join it.

161. Why does Vesa ask whether everybody can buy priority-plus?

A) She wants to purchase tickets for every passenger.

B) She recognizes that a widely purchased upgrade might recreate the same lack of relative advantage.

C) She believes priority-plus tickets are required to travel.

D) She has forgotten which sailing her ticket names.

162. In Vesa’s statement about paying for a privilege, privilege most nearly means:

A) a handwritten receipt

B) an unavoidable mistake

C) a fixed departure time

D) a special benefit

163. How do the final stage directions strengthen the scene’s irony?

A) They show no physical progress despite the language of improved priority.

B) They reveal that Vesa secretly bought another upgrade.

C) They prove the clerk has left the terminal.

D) They move the scene to a different day.

164. What pattern chiefly develops the conflict?

A) Vesa and the clerk repeatedly exchange their occupations.

B) Vesa tests formal promises against their practical effect, while the clerk keeps restating categories and offering another category.

C) The dialogue lists ferry destinations in alphabetical order.

D) The clerk explains the ferry’s engine before selling repairs.

Informational Passage: Cold to the Touch

A metal spoon and a wooden spoon left in the same room long enough can begin at the same temperature. Touch them with a warm hand, however, and the metal may feel colder. The sensation does not by itself establish that the metal started colder.

When the hand is warmer than the spoon, energy moves from the skin into the object. Metal generally conducts heat away from the contact region more readily than wood does. It can therefore keep drawing energy from the touching skin quickly, cooling that small region of skin more rapidly.

Wood transfers heat through itself more slowly. Its contact region warms sooner and limits further rapid heat loss from the skin. The hand consequently experiences a different rate of cooling, even though the two objects began at the same temperature.

This is a comparison of heat transfer, not a claim that metal is always colder than wood. If both objects start hotter than the skin, the direction of transfer reverses. The more rapid transfer can then make the metal feel hotter instead.

A thermometer and a hand are thus answering different questions. A suitable thermometer can compare the objects' initial temperatures. The hand registers the effect of their interaction with the skin, including how quickly energy moves. Feeling colder and having a lower initial temperature are related possibilities, but they are not interchangeable descriptions.

165. What starting condition allows the two spoons to have the same initial temperature in the passage?

A) One has just been removed from an oven.

B) The wooden spoon is twice as long.

C) They have remained in the same room long enough.

D) Both have been held in a hand continuously.

166. What distinction is central to the explanation?

A) A wooden spoon contains no material.

B) Every cold sensation proves a lower initial temperature.

C) Metal always has a lower temperature than nearby wood.

D) An object's initial temperature differs from the rate at which it transfers heat during touch.

167. Two objects begin equally HOTTER than the skin. According to the passage, what would more rapid heat transfer tend to do?

A) Make the object lose all ability to transfer heat

B) Prove that the skin was initially hotter than both objects

C) Make the faster-transferring object feel hotter rather than colder

D) Guarantee that the slower-transferring object has no temperature

168. In the final paragraph, registers most nearly means:

A) enrolls in a course

B) detects or responds to

C) pays at a counter

D) changes the object's identity

169. Why does the author include the case of objects hotter than the skin?

A) To replace the main explanation with a recipe

B) To prove that thermometers cannot compare temperatures

C) To claim that wood and metal always feel identical

D) To limit the explanation to heat-transfer direction rather than treating coldness as a permanent property of metal

Humanities Passage: A Street Built for One View

Imagine a theater set that suggests a long street within a shallow stage. The doorway nearest the audience is nearly full size. Successive doorways are smaller, and the gaps between them shrink. Floor lines narrow toward a point farther back. From a chosen viewing position, these changes can make the street seem to extend much farther than it physically does.

The design borrows expectations from ordinary seeing. Distant buildings appear smaller, and parallel road edges seem to approach one another. The set exaggerates and arranges those cues within a restricted space. Its apparent depth is constructed from relationships among the forms, not from a hidden stretch of actual street.

This illusion depends partly on where the observer sits. From far to one side, edges that align from the intended position may separate, exposing the arrangement. The stage street is foreshortened: its physical depth is compressed in order to suggest a greater depth from that selected view.

An audience can understand this construction without losing all pleasure in it. Watching a play does not require believing that a real town has been fitted behind the curtain. The painted and built forms serve a dramatic purpose. Judging them only by the standards of a street that people must use every day would miss the particular achievement: making a limited space support a convincing imagined one.

170. Which changes in the successive doorways help create the street illusion?

A) Every doorway is made larger than the one before it.

B) The floor is extended into an actual hidden town.

C) The doorways and the gaps between them become smaller farther back.

D) The gaps grow while all doorway sizes remain identical.

171. What chiefly motivates the discussion of the shallow stage and its apparent street?

A) To provide building regulations for public streets

B) To prove that audiences must mistake every stage set for reality

C) To rank theaters by their actual street lengths

D) To explain how a viewpoint-dependent design can create theatrical depth and why it should be judged by its dramatic purpose

172. What does the passage suggest about a spectator seated far outside the intended viewing position?

A) The spectator must see a longer real street.

B) The spectator may see separations that make the set’s physical construction more apparent.

C) The spectator’s position cannot affect apparent alignment.

D) The stage must physically expand to restore the illusion.

173. As defined in the third paragraph, foreshortened means:

A) physically compressed in depth while suggesting greater depth from a selected view

B) made wider than every nearby building

C) hidden from every possible observer

D) repaired with stronger materials

174. How does "a real town has been fitted behind the curtain" function in the argument?

A) It presents an exaggerated literal belief that the audience need not hold to appreciate the illusion.

B) It describes a construction project the author says actually occurred.

C) It supplies an exact measurement of the stage.

D) It proves that every spectator rejects imaginative settings.

Section 4: Mathematics

64 questions | Recommended time: 45 minutes

175. A point lies 3/8 unit to the RIGHT of −4 on a number line. Which number labels that point?

A) −4 3/8

B) −3 3/8

C) −3 5/8

D) 3 5/8

176. Which choice lists ALL fractions n/48, with integer n, that lie STRICTLY between 1/4 and 1/3?

A) 13/48, 14/48, and 15/48 only

B) 13/48 and 15/48 only

C) 12/48, 13/48, 14/48, and 15/48 only

D) 12/48, 13/48, 14/48, 15/48, and 16/48 only

177. How many zeros occur at the END of the integer product 1 × 2 × 3 × 4 × 5 × 6 × 7 × 8 × 9 × 10?

A) No trailing zeros

B) One trailing zero

C) Two trailing zeros

D) Four trailing zeros

178. Which choice gives ALL positive integers n satisfying 0.08 < 1/n < 0.10?

A) 10 and 11 only

B) 11 and 12 only

C) 12 and 13 only

D) 10, 11, and 12 only

179. For t minutes after a reading begins, an instrument shows C = 90 − 8t. What does the 8 represent?

A) The initial reading is 8 units.

B) The reading increases by 8 units per minute.

C) The reading reaches zero after exactly 8 minutes.

D) The reading decreases by 8 units per minute.

180. A solution to (x − 2)/5 = (x + 1)/8 begins incorrectly with 5(x − 2) = 8(x + 1). Which equation correctly replaces that FIRST step?

A) 5(x + 2) = 8(x − 1)

B) 8x − 2 = 5x + 1

C) 8(x − 2) = 8(x + 1)

D) 8(x − 2) = 5(x + 1)

181. Two numbers a and b satisfy a + b = −3 and ab = −10. What is (a + 1)(b + 1)?

A) −13

B) −6

C) 14

D) −12

182. Which positive choice of c makes |x − 4| = c have one POSITIVE and one NEGATIVE solution?

A) c = 1

B) c = 2

C) c = 6

D) c = 4

183. A triangular prism has two triangular faces and three rectangular faces. How many edges does it have?

A) 5

B) 9

C) 6

D) 18

184. One exterior angle at each vertex of a triangle is selected, with each exterior supplementary to its interior angle. These exterior angles are in the ratio 2:3:4. What is the SMALLEST interior angle?

A) 40°

B) 20°

C) 60°

D) 80°

185. The four vertices of a rectangle are (−3, −4), (3, −4), (3, 4), and (−3, 4). A circle passes through all four. What is the circle's area?

A) 25π square units

B) 20π square units

C) 48π square units

D) 100π square units

186. A machinist changes a drawing's length label from 12 centimeters to 120 millimeters. What happens to the length specified by the drawing?

A) It remains the same.

B) It becomes ten times as long.

C) It becomes one tenth as long.

D) It increases by 108 millimeters.

187. An image is 320 pixels wide and 240 pixels high. It is printed without cropping at 80 pixels per inch in both directions. What is the printed image's perimeter?

A) 7 inches

B) 28 inches

C) 14 inches

D) 12 inches

188. For the data set 2, 4, 6, which statement is correct? A mode must occur more often than the other values.

A) The mean and median are 4, but there is no mode.

B) The mean, median, and mode are all 4.

C) The mean is 4, but the median is 6.

D) The mode is 2 because it is listed first.

189. A fair six-sided die is rolled. If it shows 6, keep that result. Otherwise, roll once more and keep the SECOND result. What is the probability that the kept result is 6?

A) 1/6

B) 1/3

C) 5/36

D) 11/36

190. An integer is chosen uniformly from 10 through 99. What is the probability that its two digits are EQUAL or that its units digit is ZERO?

A) 1/10

B) 1/5

C) 19/90

D) 2/9

191. Which integer is divisible by 4 but NOT by 8?

A) 128

B) 130

C) 136

D) 124

192. A quantity is recorded as ZERO whole units, THIRTEEN tenths, and TWO hundredths. Which ordinary decimal represents it?

A) 0.132

B) 13.02

C) 1.302

D) 1.32

193. A reduced fraction has a terminating decimal exactly when its denominator has no prime factors other than 2 or 5. What is the GREATEST integer n, with 0 < n < 72, for which n/72 terminates?

A) 71

B) 64

C) 63

D) 70

194. Which expression is equivalent to a − (b − a) for all real a and b?

A) −b

B) 2a − b

C) b

D) 2a + b

195. Which choice gives ALL integers x satisfying 3 < |x| ≤ 5?

A) Four: −5, −4, 4, and 5

B) Two: 4 and 5

C) Six: −5, −4, −3, 3, 4, and 5

D) Three: −4, 4, and 5

196. Which equation fits EVERY column of the table?
x: 2, 3, 5, 6
y: 30, 20, 12, 10

A) $y = 60 - x$

B) $y = 10x$

C) $y = x + 28$

D) $y = 60/x$

197. For which real values of x are EXACTLY TWO of these expressions positive?
$x + 2$, $x - 1$, $4 - x$

A) $1 < x < 4$

B) $-2 < x \leq 1$ or $x \geq 4$

C) $-2 < x < 1$ or $x > 4$

D) $x \leq -2$ or $1 < x < 4$

198. A triangle is translated horizontally by a NONZERO distance, without rotation or resizing. Which property must change?

A) Its side lengths

B) Its angle measures

C) Its vertex coordinates

D) Its area

199. Rotate point (2, −5) by 90° CLOCKWISE about (1, 1). What is the image?

A) (−5, −2)

B) (7, 2)

C) (−1, 7)

D) (−5, 0)

200. A square of side 6 units is cut along one diagonal into two triangles. What is the SUM of the two triangles' perimeters?

A) $24 + 6\sqrt{2}$ units

B) $24 + 12\sqrt{2}$ units

C) 24 units

D) 48 units

201. Two timers start together. One runs for 1:05, meaning 1 minute 5 seconds; the other runs for 1.5 minutes. Which finishes later, and by how much?

A) The 1:05 timer, by 25 seconds

B) The 1.5-minute timer, by 45 seconds

C) The 1.5-minute timer, by 25 seconds

D) They finish together.

202. A stack of 120 identical cards is 3.6 centimeters thick. Ignore air gaps. What is one card's thickness in MILLIMETERS?

A) 0.03 millimeter

B) 0.3 millimeter

C) 3 millimeters

D) 30 millimeters

203. An unfamiliar temperature scale is linear: 0°Z equals −20°C, and 100°Z equals 180°C. What Celsius temperature equals 25°Z?

A) 50°C

B) 5°C

C) 30°C

D) 70°C

204. For this question, percentile rank means the percentage of observations STRICTLY BELOW a value. The data are 10, 10, 12, 15, 15, 20. What is the percentile rank of 15?

A) 50%

B) 33 1/3%

C) 66 2/3%

D) 83 1/3%

205. Append a real number t to the data 1, 4, 8, 11. Which condition gives ALL and ONLY the values of t for which the new median is 8?

A) $t \geq 8$

B) $4 \leq t \leq 8$

C) $t = 8$ only

D) Every real value of t

206. Disk F has one H face and one T face. Disk D has H on BOTH faces. Choose a disk with equal probability, then reveal either face with equal probability. Given that H appears, which probability and justification correctly describe the chance you chose F?

A) 1/3: one of the three equally likely H-showing disk-face outcomes belongs to F.

B) 1/2: the disks began equally likely, and seeing H cannot change that.

C) 2/3: two H-showing outcomes belong to D, so that is the chance of F.

D) 3/4: three of all four disk-face outcomes show H.

207. In 5,050, how many times as large is the value of the LEFT 5 as the value of the RIGHT 5?

A) 10 times

B) 100 times

C) 1,000 times

D) The two values are equal.

208. Which number is CLOSEST to zero?

A) −1/5

B) 3/10

C) −2/7

D) 1/4

209. In a whole-number place-value system, the rightmost position is ones; every position to its LEFT is worth FOUR times as much. What ordinary decimal value does the numeral 103 represent?

A) 7

B) 67

C) 19

D) 103

210. Every integer square leaves remainder 0 or 1 when divided by 4. Which number CANNOT be written as a difference of two nonnegative integer squares?

A) 35

B) 48

C) 80

D) 30

211. Given $2x + 3y = 8$ and $5x + 3y = 17$, which equation results from subtracting the FIRST equation from the SECOND?

A) $3x = 9$

B) $3x + 6y = 9$

C) $7x = 25$

D) $-3x = 9$

212. If 3 to the power x equals 27 to the power $(x - 2)$, what is x?

A) $x = 2$

B) $x = 6$

C) $x = 9$

D) $x = 3$

213. Let $f(x) = 3x - 4$. Which expression equals $f(2x) - 2f(x)$ for EVERY real x?

A) 0

B) -4

C) 4

D) $3x - 4$

214. For real $x \neq 3$, solve $(x^2 - 9)/(x - 3) = 6$. Which statement gives the complete result?

A) $x = 3$ only

B) There is no real solution.

C) $x = -3$ only

D) Every real x except 3 is a solution.

215. On a straight line, ray AB starts at A and extends through B. Ray BA starts at B and extends through A. A and B are distinct. What is their overlap?

A) The segment from A to B, including both endpoints

B) The entire straight line

C) Only point A

D) Only point B

216. A quadrilateral has consecutive side lengths 7, 9, 12, and x units. A circle touches all four sides. Use the fact that opposite side-length sums are equal in such a quadrilateral. What is x?

A) 5 units

B) 10 units

C) 14 units

D) 19 units

217. A sphere inside a cube touches all SIX faces. The cube's edge is 6 units. What FRACTION of the cube's volume is occupied by the sphere? Use $V = (4/3)\pi r^3$ for a sphere.

A) $\pi/6$

B) $\pi/8$

C) $\pi/12$

D) $\pi/3$

218. For this problem, stored energy in watt-hours equals volts × amp-hours. An ideal 12-volt, 5-amp-hour battery supplies a 10-watt device with no energy loss. How long can it operate?

A) 0.5 hour

B) 12 hours

C) 60 hours

D) 6 hours

219. A channel has cross-sectional area 12 square centimeters and uniform flow speed 5 centimeters per second. Volume per second equals area × speed. Using 1 liter = 1,000 cubic centimeters, how many liters pass per MINUTE?

A) 0.06 liter

B) 0.6 liter

C) 3,600 liters

D) 3.6 liters

220. Each worker has one recorded break time. Crew A's mean is 8 minutes and Crew B's mean is 12 minutes. Which additional information is sufficient to determine the mean for ALL workers together?

A) The longest break in each crew

B) The number of workers in each crew

C) The median break in each crew

D) The starting time of each crew's shift

221. Four musicians, A, B, C, and D, take four chairs in a row in a uniformly random order. What is the probability that A and B are NOT adjacent?

A) 6/24

B) 8/24

C) 12/24

D) 18/24

222. Eight repair tags consist of TWO tags in EACH of four colors. Draw tags without looking and without replacement. What is the LEAST number that GUARANTEES at least two complete same-color pairs?

A) 4 tags

B) 5 tags

C) 6 tags

D) 8 tags

223. Add ONE pair of parentheses to -4^2 without changing any other symbol. Which placement makes the value POSITIVE?

A) $-(4^2)$

B) $-(4)^2$

C) $(-4)^2$

D) (-4^2)

224. An integer code must be divisible by BOTH 6 and 14. Which range contains NO valid code?

A) 85 through 125

B) 40 through 50

C) 80 through 90

D) 120 through 130

225. In a perfect square, every prime-factor exponent is even. Can a POSITIVE integer n make BOTH 2n and 3n perfect squares?

A) n = 6 works.

B) n = 12 works.

C) n = 18 works.

D) No positive integer n can do this.

226. A student claims 5(x + 2) = 5x + 2. Which correction fixes the distribution error?

A) Delete the 5, giving x + 2.

B) Multiply the 2 by 5 as well, giving 5x + 10.

C) Add 5 to the 2, giving 5x + 7.

D) Multiply x by 2, giving 10x + 2.

227. For all real x and y, let $U = x^2$, $V = y^2$, and $W = (x + y)^2$. Which expression ALWAYS equals xy?

A) W − U − V

B) (W − U − V)/4

C) (W − U − V)/2

D) (W + U + V)/2

228. For positive x and y with x ≠ y, simplify (1/x + 1/y) ÷ (1/x − 1/y).

A) (x + y)/(y − x)

B) (x + y)/(x − y)

C) 1

D) (y − x)/(x + y)

229. Solve ||x| − 3| = 2. Which choice gives ALL real solutions?

A) {−5, −1, 1, 5}

B) {−5, 5}

C) {−1, 1}

D) {1, 5}

230. In triangle PQR, the angle formed by QP and QR is a right angle. Which side is the hypotenuse?

A) PR

B) PQ

C) QR

D) This triangle has no hypotenuse.

231. A circle has center (3, 0) and radius 3. Reflect it across the y-axis. How many points belong to BOTH the circle and its reflected image?

A) 0 points

B) 2 points

C) Infinitely many points

D) 1 point

232. The areas of three faces meeting at one corner of a rectangular prism are 6, 10, and 15 square units. What is the prism's volume?

A) 31 cubic units

B) 60 cubic units

C) 30 cubic units

D) 900 cubic units

233. Distance is measured in kilometers and speed in kilometers per hour. What UNIT results when distance is divided by speed?

A) kilometers

B) square kilometers per hour

C) hours per kilometer

D) hours

234. A turntable makes 45 complete revolutions per minute at a steady rate. A mark starts facing the front. Where does it face after FOUR seconds?

A) 90° past its original direction

B) Its original direction

C) 180° past its original direction

D) 270° past its original direction

235. A rectangle's length and width are recorded as 8 cm and 5 cm, each rounded to the nearest whole centimeter. Which ACTUAL area is impossible?

A) 34 square centimeters

B) 40 square centimeters

C) 46 square centimeters

D) 48 square centimeters

236. Ordered times in seconds are 2, 3, 5, 8, 9, 11, 13, 20. Define Q1 as the median of the LOWER four values and Q3 as the median of the UPPER four. What is Q3 − Q1?

A) 18 seconds

B) 8 seconds

C) 4 seconds

D) 12 seconds

237. In a 12-minute demonstration, light A is on from minute 0 up to minute 4, and light B from minute 3 up to minute 9. An instant is chosen uniformly during the demonstration. What is the probability BOTH are on?

A) 1/6

B) 3/4

C) 1/12

D) 5/6

238. A and B alternate independent fair-coin tosses, with A first. The first player to toss HEADS wins. After two tails, the situation returns to A's starting turn. Which probability of A winning is correct?

A) 1/2; a fair coin makes the players equally likely to win.

B) 2/3; starting first gives A an advantage.

C) 3/4; A wins whenever either of the first two tosses is heads.

D) 1/3; starting first puts A at a disadvantage.

Section 5: Language

60 questions | Recommended time: 30 minutes

239. Which version correctly uses a hyphen in the written-out number?

A) The shipment contains twenty one brushes.

B) The shipment contains twenty, one brushes.

C) The shipment contains twenty-one brushes.

D) The shipment contains twenty:one brushes.

240. A formal directory requires a period after each personal initial and no commas within a name. Which version correctly shortens Dana Rae Vega under that rule?

A) D, R, Vega

B) D: R: Vega

C) D. R. Vega

D) D R, Vega

241. An editor preserves the quotation "Two otters was resting" and uses [sic] immediately after the word containing the original grammatical error. Which version follows that instruction?

A) "Two [sic] otters was resting."

B) "Two otters were [sic] resting."

C) "Two otters was resting [sic]."

D) "Two otters was [sic] resting."

242. A notice wraps onto a second line inside ONE sentence; / marks the line break. Which version capitalizes only words that require capitals?

A) Keep the Lid / closed during transport.

B) Keep the lid / Closed during transport.

C) Keep the lid / closed during transport.

D) Keep the lid / closed during Transport.

243. A report capitalizes numbered references such as Table 2, but uses lowercase for ordinary furniture. Which sentence follows that rule?

A) table 2 gives the dimensions of the Table beside the door.

B) Table 2 gives the dimensions of the Table beside the door.

C) table 2 gives the dimensions of the table beside the door.

D) Table 2 gives the dimensions of the table beside the door.

244. Choose the correctly spelled name of the paper-folding art.

A) oragami

B) origami

C) origamme

D) oragamme

245. Which spelling correctly names a vehicle’s supporting frame?

A) chasis

B) chassiss

C) chassus

D) chassis

246. The lamps are too bright. Which command uses the object pronoun in the correct position and form?

A) Turn them off.

B) Turn off them.

C) Turn they off.

D) Turn off they.

247. Choose the pair: "They had the porter ___ the crates and got the driver ___."

A) to carry; wait

B) carried; waiting

C) carry; to wait

D) carrying; waited

248. People now believe that the sculpture was moved last year. Complete the equivalent sentence: "The sculpture is believed ___ last year."

A) to be moving

B) to have moved

C) to be moved

D) to have been moved

249. Choose the correct pair: "___ the passengers ___ delayed."

A) Maybe; may be

B) May be; maybe

C) Maybe; maybe

D) May be; may be

250. Choose the pair that distinguishes FINANCIAL CONDITIONS from INEXPENSIVE OPERATION: "The ___ report explains trade changes; the ___ heater uses little fuel."

A) economic; economical

B) economical; economic

C) economic; economic

D) economical; economical

Editing Passage: A Useful File Name

(1) Shared files are easier to sort when their names follow a consistent pattern. (2) For a team recording weekly shoreline observations, each file name can begin with a date written in year-month-day order. (3) An ordinary alphabetical sort will then group the files by year, month, and day. (4) Months and days should each use two digits, including a leading zero when needed. (5) For example, 2026-03-07 follows this pattern. (6) ___, a name that sorts correctly can still be unhelpful if it gives no clue about the file's contents. (7) Adding a precise subject after the date can solve that problem. (8) A short agreed pattern can help the team both locate and recognize its records.

251. Which transition in sentence 6 best introduces the limitation of a date that sorts correctly?

A) For that reason

B) However

C) Similarly

D) For instance

252. Which addition to sentence 7 best illustrates its advice without introducing an unrelated naming rule?

A) Everyone should instead name every file newfile.

B) For instance, 2026-03-07-tide-notes identifies both the date and the subject.

C) Some shorelines contain more rocks than sand.

D) A file name should never contain information about its contents.

253. The writer wants to add: "Without consistent digit widths, the names may not sort in the intended order." Where would it best explain an existing instruction?

A) Immediately after sentence 4

B) Immediately before sentence 1

C) Immediately after sentence 6

D) Immediately after sentence 8

254. Which version correctly ends a sentence whose final word is the abbreviation Inc.?

A) The parts came from Rivet Inc..

B) The parts came from Rivet Inc.

C) The parts came from Rivet Inc,

D) The parts came from Rivet Inc:

255. Original: "The engine is not, despite the optimistic report, ready for testing." Which shortened quotation uses an ellipsis while preserving the original claim?

A) "The engine is ... ready for testing."

B) "The engine ... is ready for testing."

C) "The engine is not ... ready for testing."

D) "The engine is ready ... not for testing."

256. Which sentence correctly writes a letter grade and an ordinary school subject that is not a course title?

A) She earned an A in algebra.

B) She earned an a in algebra.

C) She earned an A in Algebra.

D) She earned an a in Algebra.

257. A fable treats the quality patience as a character named Patience. Which sentence correctly distinguishes that character’s NAME from the ordinary quality?

A) At dusk, patience found me without any patience.

B) At dusk, Patience found me without any Patience.

C) At dusk, patience found me without any Patience.

D) At dusk, Patience found me without any patience.

258. Choose the correctly spelled word for a journey to observe wildlife.

A) saffari

B) safary

C) saffary

D) safari

259. Which spelling correctly names nourishment or support needed to continue living?

A) sustenence

B) sustanance

C) sustenance

D) sustinance

260. Which spelling correctly completes "The corridor is lit by ___ tubes"?

A) fluorescent

B) flourescent

C) fluorescant

D) fluorecsent

261. Which phrase correctly expands MIGHT’VE in "I might’ve left the ticket in my coat"?

A) might of

B) might had

C) might have

D) might is

262. In "Running this shop requires patience," which word is the MAIN FINITE VERB?

A) Running

B) requires

C) shop

D) patience

263. Choose the formal verb form: "They covered the lamp lest its glare ___ the watcher."

A) distract

B) distracts

C) distracted

D) to distract

264. Choose the pair: "They must ___ water from the skiff before loading the ___ of hay."

A) bale; bail

B) bail; bale

C) bail; bail

D) bale; bale

265. Choose the pair: "The jeweler measured the diamond in ___; the editor drew a ___ to show where a word should be inserted."

A) carets; carat

B) carats; carat

C) carets; caret

D) carats; caret

Editing Passage: Describing a Sculpture Aloud

(1) A clear audio description should help listeners build a stable picture before adding fine details. (2) Consider a sculpture consisting of a pale sphere resting on a dark cube, with a narrow polished metal band circling the sphere. (3) A rough patch on the cube's back edge is visible only from behind. (4) The overall arrangement places the rounded form directly above the block. (5) The band tilts toward the cube's front corner. (6) It catches the light sharply, unlike the sphere's dull surface. (7) The description should also state which side is being treated as the front, so that later directions remain consistent. (8) Small features become easier to locate once the listener understands the arrangement.

266. To follow sentence 1’s stated plan, which move would most improve the paragraph’s development?

A) Move sentence 2 to follow sentence 8.

B) Move sentence 3 to follow sentence 6.

C) Move sentence 4 before sentence 1.

D) Move sentence 8 between sentences 2 and 3.

267. Which replacement for IT in sentence 6 most clearly identifies the intended shiny element?

A) The metal band

B) The sculpture

C) The dark cube

D) The pale sphere

268. Which revised final sentence best connects the paragraph’s advice to its purpose?

A) A sculpture’s price is always more useful than its shape.

B) Only people looking directly at the sculpture can understand any description.

C) Every surface detail should be listed before the sculpture’s main parts.

D) A consistent viewpoint and an overall arrangement give listeners a framework for locating smaller features.

269. Which version correctly ends a NEUTRAL POLITE COMMAND that contains an embedded question?

A) Please tell me where the entrance is?

B) Please tell me where the entrance is.

C) Please tell me where the entrance is;

D) Please tell me where the entrance is:

270. Which version correctly places the comma between the paired comparative clauses?

A) The faster, the wheel turned the louder the hum became.

B) The faster the wheel, turned the louder the hum became.

C) The faster the wheel turned, the louder the hum became.

D) The faster the wheel turned the louder, the hum became.

271. A complete parenthetical sentence follows another complete sentence. Which version correctly places both sentence-ending periods and capitals?

A) The exhibit is closed. (It reopens on Monday).

B) The exhibit is closed (It reopens on Monday.)

C) The exhibit is closed. (it reopens on Monday.)

D) The exhibit is closed. (It reopens on Monday.)

272. Which version uses conventional capitalization for the formal letter closing YOURS TRULY?

A) Yours truly,

B) Yours Truly,

C) yours Truly,

D) yours truly,

273. A house style capitalizes a COMPLETE sentence after a colon but not a following noun phrase. Which version follows both requirements?

A) Only one thing mattered: the door must stay shut. They needed one thing: a key.

B) Only one thing mattered: The door must stay shut. They needed one thing: A key.

C) Only one thing mattered: the door must stay shut. They needed one thing: A key.

D) Only one thing mattered: The door must stay shut. They needed one thing: a key.

274. Choose the correctly spelled word for a framework that supports climbing plants.

A) trelis

B) trellis

C) trelliss

D) trallis

275. Which spelling correctly names a system of signaling with positions of flags or arms?

A) semafore

B) semiphore

C) semaphore

D) semapohre

276. Choose the standard possessive form: "That illustrator is a friend of ___."

A) my

B) me

C) mine’s

D) mine

277. Which sentence means that the printer CEASED its printing activity?

A) The printer stopped to print.

B) The printer stopped printing.

C) The printer was about to start printing.

D) The printer did not stop printing.

278. Which revision correctly uses SUCH while preserving "The box was so heavy that it took two people to lift it"?

A) It was a such heavy box that it took two people to lift it.

B) It was such heavy a box that it took two people to lift it.

C) It was such a heavy box that it took two people to lift it.

D) It was a heavy such box that it took two people to lift it.

279. Choose the pair: "Light passed through the ___ curtain; tomorrow the farmers would ___ the sheep."

A) sheer; shear

B) shear; sheer

C) sheer; sheer

D) shear; shear

280. Choose the pair: "She will ___ the delicate subject while wearing a silver ___."

A) brooch; broach

B) broach; broach

C) broach; brooch

D) brooch; brooch

281. Choose the words meaning PURSUE A COURT CASE first and HARASS UNFAIRLY second: "The lawyer will ___ the case; the tyrant continues to ___ people for their beliefs."

A) prosecute; persecute

B) persecute; prosecute

C) prosecute; prosecute

D) persecute; persecute

Editing Passage: A Connection Made by Editing

(1) The order of film shots can encourage an expectation without explicitly stating it. (2) A short scene shows a baker drawing bread from an oven, followed by a cyclist hurrying around a corner. (3) More shots alternate between the cooling loaves and the approaching bicycle. (4) Viewers may assume the cyclist is rushing to collect the bread. (5) No shot has actually identified the cyclist's destination. (6) The final shot shows the cyclist delivering a letter on an entirely different street. (7) That ending overturns the expectation created by the earlier editing. (8) The scene demonstrates how viewers connect adjacent images, even when the film has not yet established that the actions belong to the same event.

282. Which title best expresses the paragraph’s focus on the relationship between editing and audience expectations?

A) An Arrival the Audience Imagines

B) How to Bake a Loaf

C) The Bicycle’s Mechanical Parts

D) Why Every Film Must End with a Letter

283. Which revision best combines sentences 6 and 7 while making their cause-and-effect connection clear?

A) The final shot shows the cyclist delivering a letter on an entirely different street because the audience’s earlier expectation was correct.

B) The final shot shows the cyclist delivering a letter on an entirely different street, thereby overturning the expectation created by the earlier editing.

C) The earlier editing overturns the final shot by showing that the cyclist collects bread from the baker.

D) The final shot shows a cyclist, and films sometimes end on different streets.

284. Which version uses the standard end mark for an ENTHUSIASTIC EXCLAMATION?

A) How bright the lantern is?

B) How bright the lantern is;

C) How bright the lantern is:

D) How bright the lantern is!

285. Which version avoids incorrectly separating the whole SUBJECT from its main verb?

A) What the instrument measures is pressure.

B) What the instrument measures, is pressure.

C) What the instrument measures; is pressure.

D) What, the instrument measures is pressure.

286. ONE spoken sentence is interrupted by a speech tag. Which version preserves correct capitalization?

A) "the bridge," Alban said, "will open at noon."

B) "The bridge," Alban said, "will open at noon."

C) "The bridge," Alban said, "Will open at noon."

D) "the bridge," Alban said, "Will open at noon."

287. A publisher keeps the acronym DIY uppercase, lowercases internal TO, and capitalizes BOTH parts of major hyphenated compounds. Which title follows all three requirements?

A) A Diy Guide to Heat-Resistant Coatings

B) A DIY Guide To Heat-Resistant Coatings

C) A DIY Guide to Heat-Resistant Coatings

D) A DIY Guide to Heat-resistant Coatings

288. Which spelling correctly names a rope used as a lasso?

A) lariat

B) lairat

C) larriat

D) lariatt

289. Choose the correctly spelled word for a fine writing material traditionally prepared from animal skin.

A) velumm

B) vellom

C) vellum

D) vellem

290. Which spelling correctly names time available for rest or recreation?

A) liesure

B) leisure

C) leizure

D) leisuer

291. Replace THE CHILD with a pronoun while preserving LOOK AFTER, meaning care for. Which sentence is standard?

A) The volunteer will look him after.

B) The volunteer will look after he.

C) The volunteer will look he after.

D) The volunteer will look after him.

292. Which sentence places ALMOST so it limits HOW MANY visitors signed, rather than how near they came to signing?

A) All visitors almost signed the book.

B) Almost all visitors signed the book.

C) All visitors signed almost the book.

D) All visitors signed the book almost.

293. Choose the standard American English completion: "Beneath the rafter ___ a row of small lamps."

A) hang

B) are hanging

C) hangs

D) have hung

294. Choose the words meaning GOVERNMENT REPRESENTATIVE ABROAD first and COMFORT second: "The ___ offered to ___ the distressed traveler."

A) console; consul

B) consul; consul

C) console; console

D) consul; console

295. Choose the word for a BEAR first and the word meaning HORRIFYING second: "The photographer described a ___ bear, while the storyteller told a ___ tale."

A) grisly; grizzly

B) grizzly; grizzly

C) grizzly; grisly

D) grisly; grisly

296. Choose the pair: "The treasurer will ___ the funds, and the crowd will ___ after the concert."

A) disburse; disperse

B) disperse; disburse

C) disburse; disburse

D) disperse; disperse

Editing Passage: Connections for a Working Model

(1) The best connection for a model depends on what the model must do at its present stage. (2) An architect comparing several layouts needs to move small model walls and doorways repeatedly. (3) Permanent glue can keep the pieces firm, but removing them may damage parts that are still useful. (4) Removable clips can hold the pieces while allowing them to be repositioned. (5) The clips may show in photographs, yet that drawback may matter less during discussion than the ability to revise a layout. (6) Frequent changes make reversibility especially valuable. (7) ___, hidden attachments may be better for the display version. (8) The choice should follow the model's current purpose rather than a rule that one fastening method is always best.

297. Which added detail would most directly support sentence 6?

A) During one design discussion, the doorway may need to be moved to several alternative positions.

B) The final display label will be printed in dark blue.

C) A permanent connection is useful because it can never be adjusted.

D) The architect has decided that no layout changes will be considered.

298. Which clause best begins sentence 7 by marking the stage change that justifies a different fastening choice?

A) Before any alternatives have been compared

B) Because clips make revision impossible

C) Even though display models are never shown

D) Once the layout is settled

Answer Key

Question numbers continue across all five sections.

Verbal Skills

1. C	2. D	3. B	4. C	5. B	6. B	7. C	8. A
9. B	10. C	11. A	12. A	13. D	14. A	15. D	16. B
17. C	18. B	19. B	20. A	21. C	22. D	23. D	24. C
25. A	26. B	27. D	28. A	29. C	30. D	31. D	32. A
33. D	34. C	35. B	36. C	37. D	38. A	39. C	40. B
41. C	42. A	43. B	44. D	45. A	46. B	47. C	48. D
49. D	50. A	51. B	52. A	53. D	54. B	55. C	56. C
57. C	58. D	59. A	60. B				

Quantitative Skills

61. B	62. A	63. B	64. C	65. D	66. A	67. B	68. A
69. D	70. C	71. B	72. C	73. D	74. C	75. B	76. A
77. D	78. C	79. A	80. B	81. A	82. D	83. C	84. B
85. C	86. D	87. D	88. C	89. D	90. A	91. B	92. C
93. D	94. A	95. D	96. C	97. B	98. A	99. B	100. A
101. D	102. B	103. A	104. C	105. D	106. A	107. A	108. D
109. C	110. B	111. C	112. B				

Reading Comprehension

113. D	114. B	115. D	116. C	117. B	118. A	119. A	120. D
121. C	122. C	123. A	124. C	125. B	126. B	127. D	128. A
129. D	130. B	131. A	132. B	133. C	134. A	135. C	136. D
137. C	138. A	139. D	140. B	141. B	142. C	143. A	144. D
145. D	146. A	147. D	148. A	149. C	150. D	151. B	152. B
153. D	154. C	155. B	156. A	157. A	158. C	159. B	160. C
161. B	162. D	163. A	164. B	165. C	166. D	167. C	168. B
169. D	170. C	171. D	172. B	173. A	174. A		

Mathematics

175. C	176. A	177. C	178. B	179. D	180. D	181. D	182. C
183. B	184. B	185. A	186. A	187. C	188. A	189. D	190. B
191. D	192. D	193. C	194. B	195. A	196. D	197. B	198. C
199. D	200. B	201. C	202. B	203. C	204. A	205. A	206. A

207. B	208. A	209. C	210. D	211. A	212. D	213. C	214. B
215. A	216. B	217. A	218. D	219. D	220. B	221. C	222. C
223. C	224. A	225. D	226. B	227. C	228. A	229. A	230. A
231. D	232. C	233. D	234. B	235. D	236. B	237. C	238. B

Language

239. C	240. C	241. D	242. C	243. D	244. B	245. D	246. A
247. C	248. D	249. A	250. A	251. B	252. B	253. A	254. B
255. C	256. A	257. D	258. D	259. C	260. A	261. C	262. B
263. A	264. B	265. D	266. B	267. A	268. D	269. B	270. C
271. D	272. A	273. D	274. B	275. C	276. D	277. B	278. C
279. A	280. C	281. A	282. A	283. B	284. D	285. A	286. B
287. C	288. A	289. C	290. B	291. D	292. B	293. C	294. D
295. C	296. A	297. A	298. D				

Detailed Explanations

Verbal Skills

1. HSPT-V10-001 - Answer C: abundant
Copious describes something present in large amounts; abundant preserves that meaning, while the other choices concern strength, certainty, or distance.

2. HSPT-V10-002 - Answer D: open to considering new ideas
Receptive describes willingness to receive and consider suggestions or information; it does not imply interruption, deafness, or resistance to change.

3. HSPT-V10-003 - Answer B: difficult to understand
Abstruse means difficult to comprehend, usually because of complexity or obscurity; neither brevity, humor, nor numerical content defines it.

4. HSPT-V10-004 - Answer C: contented
Wistful suggests a gently sad longing for something absent; contented instead describes satisfaction with what one has.

5. HSPT-V10-005 - Answer B: careless
Scrupulous can describe careful attention to correctness and ethical standards; careless reverses that attention rather than reinforcing it.

6. HSPT-V10-006 - Answer B: permeable to water
Impervious material does not let water pass through; permeable material does, making permeability the relevant opposite.

7. HSPT-V10-007 - Answer C: A rack-and-pinion mechanism that moves a straight toothed bar when its gear turns
Both mechanisms convert rotation into straight movement; the alternatives reflect light, protect an object, or convert airflow into sound.

8. HSPT-V10-008 - Answer A: Shuffling a deck changes card order without adding or removing cards.
An anagram and a shuffle rearrange the same individual elements without changing their identities or quantities; the alternatives remove, combine, or resize material.

9. HSPT-V10-009 - Answer B: A cookie cutter defines a shape but supplies none of the dough.
The stencil and cutter constrain a product’s shape without supplying its material; the alternatives contain the substance they use or release.

10. HSPT-V10-010 - Answer C: bicolored
Bicolored means having two colors; its bi- carries the same two meaning, unlike the coincidental initial letters in the alternatives.

11. HSPT-V10-011 - Answer A: the backbone of an argument
Backbone figuratively names the argument’s essential supporting structure; the other phrases identify actual anatomical parts of people.

12. HSPT-V10-012 - Answer A: draft, sketch, rough cut
A draft, sketch, and rough cut can each be a preliminary version awaiting development; the alternatives include final outcomes or spatial terms.

13. HSPT-V10-013 - Answer D: The bouquet is exempt from this delivery restriction.
The stated exception covers flowers intended for the stage; it does not establish access elsewhere or the carrier’s occupation.

14. HSPT-V10-014 - Answer A: Two catalog entries need not represent two different works.
The entries are different descriptions of one identified work, so counting entries alone would overcount distinct works in this case.

15. HSPT-V10-015 - Answer D: At least one word is printed in black.
A word without an underline cannot be red under the rule; since black and red exhaust the possibilities, that word must be black.

16. HSPT-V10-016 - Answer B: hold and use it
Wield means hold and use something, especially a tool or weapon; concealing, repairing, and exchanging do not express that action.

17. HSPT-V10-017 - Answer C: remove less suitable choices
Winnow means separate out unwanted material; applied to proposals, it describes narrowing the list by removing less suitable candidates.

18. HSPT-V10-018 - Answer B: refined and politely behaved
Genteel describes polished manners and social refinement; the alternatives instead suggest haste, distrust, or forceful demands.

19. HSPT-V10-019 - Answer B: unfit for cultivation
Arable concerns suitability for cultivation; land unfit for cultivation reverses that meaning, while the alternatives preserve it.

20. HSPT-V10-020 - Answer A: agreeable and cooperative
Contentious means inclined to argue or quarrel; an agreeable, cooperative participant displays the opposite approach to disagreement.

21. HSPT-V10-021 - Answer C: plain and restrained
Florid style is elaborately ornamented; plain, restrained language removes that ornamentation rather than adding more of it.

22. HSPT-V10-022 - Answer D: Evaporation changes liquid water into vapor.
Evaporation reverses the vapor-to-liquid change by taking liquid water back to vapor; the alternatives perform different transformations.

23. HSPT-V10-023 - Answer D: A junction where separate traffic routes merge
Both a confluence and the described junction are meeting points where separate paths combine; height, storage, and overlooking are different relationships.

24. HSPT-V10-024 - Answer C: An ambiguous drawing that can be seen as either a vase or two faces
The pun and ambiguous drawing preserve the same presented form while allowing different interpretations; the alternatives change size, cover damage, or remove information.

25. HSPT-V10-025 - Answer A: always
Always describes frequency, like never, seldom, and often; nearby, abroad, and backward instead indicate place or direction.

26. HSPT-V10-026 - Answer B: sell
Selling transfers ownership to the buyer; hiring, borrowing, and renting concern use or possession without that purchase-based transfer.

27. HSPT-V10-027 - Answer D: oscillate, shuttle, seesaw
Oscillate, shuttle, and seesaw all describe movement between positions in alternating directions; the other groups include one-way changes or remaining still.

28. HSPT-V10-028 - Answer A: Visitors have permission to request a tour.
May grants permission; it does not create an obligation, establish a price, or add a condition about arriving together.

29. HSPT-V10-029 - Answer C: The prediction was written before the predicted event.
Accuracy concerns agreement with the event; without evidence of the writing date, it does not establish that the statement preceded the event.

30. HSPT-V10-030 - Answer D: It was inspected and not signed.
An unsealed form cannot be signed because signing requires sealing; acceptance then requires inspection as the other exclusive possibility.

31. HSPT-V10-031 - Answer D: provoke that person
Goad means provoke or urge into action; the taunt seeks a response rather than offering comfort, information, or protection.

32. HSPT-V10-032 - Answer A: blocks its progress
Stymie means obstruct or frustrate progress; an obstacle that stymies an effort prevents it from advancing as intended.

33. HSPT-V10-033 - Answer D: calm and showing little emotion
Stolid describes an unemotional, impassive manner; restlessness, impulsiveness, and dramatic expression contrast with that steadiness.

34. HSPT-V10-034 - Answer C: vigorous
Languid suggests low energy or sluggishness; vigorous indicates energetic, forceful movement rather than continued listlessness or ease.

35. HSPT-V10-035 - Answer B: fresh and original
Trite means unoriginal through overuse; a fresh, original remark contrasts with the familiar expressions described by the other choices.

36. HSPT-V10-036 - Answer C: abstemious
Voracious describes consuming eagerly or in large amounts; abstemious describes restraint in eating and drinking, reversing that tendency.

37. HSPT-V10-037 - Answer D: A paragraph break divides existing text without adding words.
Both changes introduce an internal division while leaving the amount of underlying content unchanged; adding, translating, and duplicating perform different operations.

38. HSPT-V10-038 - Answer A: A contrail marks part of a plane’s route after the plane has passed.
The wake and contrail are lingering traces of an object’s past movement; the alternatives indicate a current direction or location.

39. HSPT-V10-039 - Answer C: A border crossing is a departure from one country and an arrival in the other.
Expense and income describe the same transferred payment from opposite sides, just as departure and arrival describe one crossing from its two endpoints.

40. HSPT-V10-040 - Answer B: squeeze
Squeeze uses pressure from the hands, like kneading, wringing, or clenching; the other choices concern hearing, looking, or producing a sound.

41. HSPT-V10-041 - Answer C: is a sibling of
If one person is a sibling of another, the reverse relationship also holds; parenthood, greater age, and supervision do not reverse that way.

42. HSPT-V10-042 - Answer A: equal/unequal; complete/incomplete; occupied/unoccupied
Unequal, incomplete, and unoccupied deny the corresponding property; the other groups present degrees, roles with associated things, or related objects.

43. HSPT-V10-043 - Answer B: Reviewing every proposal does not require approving every proposal.
Consideration is the review process, while approval is one possible outcome; rejecting a reviewed proposal does not violate the promise to consider it.

44. HSPT-V10-044 - Answer D: Each unfamiliar term is explained only through the other unfamiliar term.
The definitions form a circle without an independent explanation or example, so a reader who knows neither term gains no starting meaning.

45. HSPT-V10-045 - Answer A: Neither assignment consistently fits the statement’s own claim.
If true, its claim makes it false; if false, its claim makes it true. Neither assignment remains consistent under the stated rule.

46. HSPT-V10-046 - Answer B: condemn it openly
Decry means express strong public disapproval; neutral description, enthusiastic approval, and concealment do not communicate that condemnation.

47. HSPT-V10-047 - Answer C: adjust it to fit those conditions
Attune means bring into harmony or adjustment; fitting the plan to local conditions expresses that relationship rather than announcing, repeating, or rejecting.

48. HSPT-V10-048 - Answer D: regretful
Rueful expresses regret, sometimes with a wry acceptance of one’s own mistake; the delay she caused supplies that regretful context.

49. HSPT-V10-049 - Answer D: quiet and subdued
Raucous describes harsh, unpleasant loudness; quiet, subdued voices reverse that quality rather than restating its force or noise.

50. HSPT-V10-050 - Answer A: honorable
Sordid here means morally disgraceful or dishonorable; honorable reverses that ethical description, while the other choices reinforce it.

51. HSPT-V10-051 - Answer B: guileless
Wily means skilled in deceptive tricks; guileless describes someone without such deceit, whereas cunning, crafty, and sly retain it.

52. HSPT-V10-052 - Answer A: A steady metronome’s beats mark equal time intervals.
Ruler marks and steady beats divide their respective dimensions into equal intervals; a single alarm, protective case, or cover illustration does not.

53. HSPT-V10-053 - Answer D: A ramp provides a continuous incline, while a staircase provides separate standing levels.
The matched contrast is continuous adjustment versus distinct levels; merely changing slope, rung count, or color does not reproduce that difference.

54. HSPT-V10-054 - Answer B: A reservoir’s water level stays constant because incoming water replaces outgoing water.
An unchanged total can conceal replacement of individual contents; the room and reservoir illustrate this, unlike no replacement or a visibly changing total.

55. HSPT-V10-055 - Answer C: cubby
A cubby is a small compartment used for storage; hinges, handles, and latches are fittings rather than spaces that hold contents.

56. HSPT-V10-056 - Answer C: however, nevertheless, nonetheless
However, nevertheless, and nonetheless can introduce contrast or concession; the other groups primarily concern reasons, sequence, or location.

57. HSPT-V10-057 - Answer C: sew, staple, glue
Sewing adds thread, stapling adds staples, and gluing adds adhesive; the other groups reshape, remove, or rearrange material without that fastening addition.

58. HSPT-V10-058 - Answer D: They judge routes by different criteria.
A route can minimize transfers without minimizing elapsed time; different stated criteria can therefore produce different recommendations without logical conflict.

59. HSPT-V10-059 - Answer A: Being wrong does not by itself establish knowingly saying something false.
A mistaken prediction and intentional deception differ in what the speaker believed; the later error alone does not establish deliberate falsehood.

60. HSPT-V10-060 - Answer B: The selection ensures favorable answers from those invited but does not establish everyone else’s preference.
Selecting known supporters makes their agreement unsurprising; it supplies no basis for extending the preference to engineers who were excluded.

Quantitative Skills

61. HSPT-Q10-001 - Answer B: 9
The ninth square is $9^2 = 81$; adding its digits gives 8 + 1 = 9, rather than recording the square itself.

62. HSPT-Q10-002 - Answer A: 5/17
The next numerator is 5, and the next denominator increase is 5; adding it to 12 gives 17.

63. HSPT-Q10-003 - Answer B: 64
The step after reversing 37 to 73 is subtraction of 9, so the next value is $73 - 9 = 64$.

64. HSPT-Q10-004 - Answer C: (16, 8)
From the old pair (12, 4), the sum is 16 and the difference is 8; both must use the same old pair.

65. HSPT-Q10-005 - Answer D: 70
Grouping the middle sum gives $2 \times (3 + 4) \times 5 = 70$, exceeding the results from other valid groupings.

66. HSPT-Q10-006 - Answer A: 4
An odd quotient requires removing both factors of 2 from $36 = 4 \times 9$; k = 4 is the smallest divisor that does so.

67. HSPT-Q10-007 - Answer B: Three: 37/6, 29/10, and 25/12
Swapping a and b leaves the sum unchanged. The pairs {1, 6}, {2, 5}, and {3, 4} give three distinct results.

68. HSPT-Q10-008 - Answer A: Quantity A is greater.
The greatest remainder is 7, while the positive factors of 8 are 1, 2, 4, and 8, giving four factors.

69. HSPT-Q10-009 - Answer D: The relationship cannot be determined from the information given.
When k = 2, the quantities are 2 and −2; when k = −2, both equal 2. The relationship therefore varies.

70. HSPT-Q10-010 - Answer C: The two quantities are equal.
Write $a = 7m + 5$ and $b = 7n + 6$. Every product term except 30 is divisible by 7; 30 leaves remainder 2.

71. HSPT-Q10-011 - Answer B: Quantity B is greater.
The angles sum to 180°; an angle greater than 90° leaves an angle less than 90°, so the other angle is smaller.

72. HSPT-Q10-012 - Answer C: The two quantities are equal.
A hexagon vertex connects diagonally to three nonadjacent vertices; a pentagon splits into three triangles from one vertex, making the counts equal.

73. HSPT-Q10-013 - Answer D: The relationship cannot be determined from the information given.
An axis-aligned rectangle has perimeter 12; a square with the same diagonal has perimeter $4\sqrt{10}$, greater than 12. The relationship is not fixed.

74. HSPT-Q10-014 - Answer C: 100001
The displayed numerals contain zero, one, two, and three internal zeros; the next needs four zeros between the two 1s.

75. HSPT-Q10-015 - Answer B: 2: it is the greatest factor shared by 10 and 12.
Entry 10 uses 10 and 12; their common positive factors are 1 and 2, so the greatest is 2.

76. HSPT-Q10-016 - Answer A: 3
The first eight entries reverse to 1, 2, 3, 2, 1, 2, 1, 0; adding 1 makes the next four 2, 3, 4, 3.

77. HSPT-Q10-017 - Answer D: n − 1
Subtracting 1 moves any odd integer down exactly one step to an even integer; doubling does not generally give the adjacent integer.

78. HSPT-Q10-018 - Answer C: 85
The tens digit contributes 4 ten times, or 40; the units digits sum to 0 + 1 + ... + 9 = 45, giving 85.

79. HSPT-Q10-019 - Answer A: Every arrangement leaves remainder 1 when divided by 3.
Every arrangement has digit sum 10, which leaves remainder 1 upon division by 3; changing digit order does not change that remainder.

80. HSPT-Q10-020 - Answer B: 44712
The digit sum is 13 + a + b = 16 + 2b. Its only multiple of 9 for b from 0 through 6 is 18, giving b = 1.

81. HSPT-Q10-021 - Answer A: Quantity A is greater.
There are four digit positions after the decimal point, but only the digit 8 is nonzero; four is greater than one.

82. HSPT-Q10-022 - Answer D: The relationship cannot be determined from the information given.
The pair 3 and 4 has maximum 4, while 1 and 8 has maximum 8; both means are less than 5.

83. HSPT-Q10-023 - Answer C: The two quantities are equal.
The expansion is $2x^2 + 9x + 4$; the coefficients sum to 2 + 9 + 4 = 15.

84. HSPT-Q10-024 - Answer B: Quantity B is greater.
Each diagonal is shorter than either two-side route between its endpoints, so 2d < P and 2e < P; adding gives d + e < P.

85. HSPT-Q10-025 - Answer C: The two quantities are equal.
The pentagon matches after turns of 72°, 144°, 216°, and 288°; a square has two diagonal and two midline reflection axes.

86. HSPT-Q10-026 - Answer D: The relationship cannot be determined from the information given.
The diagonal product is 72. Diagonals 6 and 12 give sum 18; diagonals 4 and 18 give sum 22, so the relationship varies.

87. HSPT-Q10-027 - Answer D: 1/5
The next multiplier is 4/5; multiplying the current 1/4 by 4/5 cancels the fours and gives 1/5.

88. HSPT-Q10-028 - Answer C: 153
The next increase is 24 × 5 = 120; adding it to 33 gives 153 rather than reporting the increase alone.

89. HSPT-Q10-029 - Answer D: 3: count the distinct prime factor 2 once, then add 1.
The distinct prime factors of 4 consist only of 2, counted once; adding 1 gives the next term, 3.

90. HSPT-Q10-030 - Answer A: 21/64
After midpoint 11/32, the current endpoints are 5/16 and 3/8. Its left half has endpoints 5/16 and 11/32, whose midpoint is 21/64.

91. HSPT-Q10-031 - Answer B: Replacing each number with its square
Squaring gives 9 for either input; addition, doubling, and sign reversal keep the two outputs different from each other.

92. HSPT-Q10-032 - Answer C: positive
The retained negative factors are −5, −4, −3, and −1, an even number; there is no zero factor, so the product is positive.

93. HSPT-Q10-033 - Answer D: 3
A reduced denominator must divide both multipliers, so it divides their greatest common factor, 3; excluding an integer rules out denominator 1.

94. HSPT-Q10-034 - Answer A: Quantity A is greater.
Distinct integers differ by at least 1, so the distance between them is at least 1 and therefore greater than 1/2.

95. HSPT-Q10-035 - Answer D: The relationship cannot be determined from the information given.
Faces 1 and 2 give odd total 3 and maximum 2; faces 5 and 6 give odd total 11 and maximum 6, reversing the comparison.

96. HSPT-Q10-036 - Answer C: The two quantities are equal.
Let x = 0.249999... . Then 100x − 10x = 24.9999... − 2.4999... = 22.5, so x = 22.5/90 = 1/4.

97. HSPT-Q10-037 - Answer B: Quantity B is greater.
Every original vertex belongs to one piece. Two new cut endpoints each appear on both pieces, adding four vertex occurrences, fewer than five.

98. HSPT-Q10-038 - Answer A: Quantity A is greater.
The hexagon has side 2 and six equilateral triangles, giving area $6\sqrt{3}$. The square has area 9; since $\sqrt{3} > 1.5$, the hexagon's area is greater.

99. HSPT-Q10-039 - Answer B: Quantity B is greater.
The exterior angle ACD equals the sum of interior angles CAB and ABC; angle ABC is positive, making ACD greater than CAB.

100. HSPT-Q10-040 - Answer A: (6, 4)
The next positive factor after 4 is 6, and 24 ÷ 6 = 4; factor 8 appears only after that pair.

101. HSPT-Q10-041 - Answer D: 24
The multiples of 8 surrounding 20 are 16 and 24; 24 is the smallest that reaches or exceeds the threshold.

102. HSPT-Q10-042 - Answer B: 7 factors of 2
The product through 7 contains four factors of 2; the next factor, $8 = 2^3$, contributes three more, bringing the count to seven.

103. HSPT-Q10-043 - Answer A: 14
Reconstruct the dividend as divisor × quotient + remainder: 4 × 3 + 2 = 14; concatenating the two recorded values would be incorrect.

104. HSPT-Q10-044 - Answer C: 2 and 2
For 2 and 2, both the sum and the product are 4; each alternative produces different values for those operations.

105. HSPT-Q10-045 - Answer D: 9/13
The first step gives (6 − 1)/(8 − 1) = 5/7; repeating on that fraction gives (10 − 1)/(14 − 1) = 9/13.

106. HSPT-Q10-046 - Answer A: Two expressions
The all-plus total is 15, so negatively signed numbers must sum to 7. Only {2, 5} and {3, 4} work, giving two sign patterns.

107. HSPT-Q10-047 - Answer A: Quantity A is greater.
Writing the maximum as $p > 0$ and minimum as $-q < 0$, the range is $p + q$, greater than either p or q.

108. HSPT-Q10-048 - Answer D: The relationship cannot be determined from the information given.
The palindrome 282 has first digit below its middle digit; 525 has first digit above its middle digit. Both digit sums are 12.

109. HSPT-Q10-049 - Answer C: The two quantities are equal.
An odd divisor of $2n$ cannot use the extra factor 2, so it already divides n; both numbers therefore have exactly the same odd divisors.

110. HSPT-Q10-050 - Answer B: Quantity B is greater.
A reflection symmetry of a triangle would exchange two vertices and force two corresponding sides equal; a triangle with all sides unequal has none.

111. HSPT-Q10-051 - Answer C: The two quantities are equal.
The squares contribute 20 edges before joining; four shared sides each remove two boundary edges, leaving $20 - 8 = 12$.

112. HSPT-Q10-052 - Answer B: Quantity B is greater.
The final length equals the original, but width and height are each half as large; final volume is therefore one quarter of the original.

Reading Comprehension

113. HSPT-R10-001 - Answer D: More help with pedaling
As the uphill stretch slows their pedals, Aunt Mina calls for more help, and the narrator responds by pushing harder.

114. HSPT-R10-002 - Answer B: The fixed rear handlebar makes the narrator mistake lack of steering control for lack of usefulness.
The comparison follows the discovery that only Mina steers; the later climb reveals a useful contribution through pedaling rather than steering.

115. HSPT-R10-003 - Answer D: present for appearance rather than practical use
The narrator feels the rear position merely looks important; the climb later disproves the belief that it serves no practical purpose.

116. HSPT-R10-004 - Answer C: It humorously conveys a feeling of being carried without contributing.
The comic image treats the narrator as luggage, emphasizing the mistaken feeling of passivity before the uphill effort demonstrates a contribution.

117. HSPT-R10-005 - Answer B: The handlebar remains unchanged, but the narrator acts on a new understanding of the rear rider's role.
The repeated steering test confirms no mechanical change; pushing before being asked shows that the narrator's understanding and participation have changed.

118. HSPT-R10-006 - Answer A: It exposes more sugar surface to the water.
The first paragraph identifies increased exposed surface as the reason water can contact more sugar at once.

119. HSPT-R10-007 - Answer A: The quantity of water and its temperature
The second paragraph expressly ties the limit to a given water quantity at a fixed temperature, rather than to the crystal's original shape.

120. HSPT-R10-008 - Answer D: A change can speed dissolving without increasing how much can remain dissolved.
The passage distinguishes the rate affected by grain size from the amount limited by water quantity and temperature.

121. HSPT-R10-009 - Answer C: has reached its dissolving limit under the stated conditions
The third paragraph defines saturation through the limit introduced earlier, not through a particular crystal size or the absence of water.

122. HSPT-R10-010 - Answer C: The finer starting grains alone do not justify a higher final dissolved amount.
The passage assigns grain size a rate effect but denies that crushing itself increases the final dissolving limit under fixed conditions.

123. HSPT-R10-011 - Answer A: Names placed along a low wall within close reach
The opening paragraph describes the second design as a low wall bearing travelers' names that visitors can approach closely.

124. HSPT-R10-012 - Answer C: To explain how monument design guides visitors' attention and participation
The comparison examines height, arrangement, and visitor movement, while explicitly allowing different personal responses to either design.

125. HSPT-R10-013 - Answer B: It directs comparison toward arrangement and experience rather than material differences.
Holding the material constant helps isolate the effects of design choices; equal materials do not require equal shapes or equal reactions.

126. HSPT-R10-014 - Answer B: especially noticeable or important
The concluding questions concern what each design brings to visitors' attention, so prominent means noticeable or important rather than concealed.

127. HSPT-R10-015 - Answer D: Its arrangement proposes movements and ways of attending, much as a script guides action.
The metaphor treats physical arrangement as guidance for behavior; the next clause expressly denies that visitors must follow it exactly.

128. HSPT-R10-016 - Answer A: To qualify the claim that design guides attention by showing that it does not control every response
The examples limit an overly strong interpretation: design invites certain encounters, while visitors can still react in different ways.

129. HSPT-R10-017 - Answer D: Tiny foxes inside the canopy
The opening identifies tiny foxes painted inside the canopy; the later opening of the umbrella reveals them among green leaves.

130. HSPT-R10-018 - Answer B: She points out that the narrator’s treatment of the gift defeats its ordinary purpose.
She has already asked whether it opens; her reply exposes the oddity of protecting a rain shelter while its owner gets soaked.

131. HSPT-R10-019 - Answer A: doing under the circumstances
The narrator contrasts the umbrella’s dry condition with getting soaked, so faring concerns how things are going rather than paying or handling objects.

132. HSPT-R10-020 - Answer B: The narrator gets soaked while keeping an umbrella dry.
The object meant to protect its owner from rain instead receives the owner’s protection, reversing the practical purpose expected of it.

133. HSPT-R10-021 - Answer C: A useful ordinary moment becomes a worthy occasion for enjoying the gift.
Opening the umbrella provides both shelter and a view of the foxes, resolving the earlier belief that ordinary use would be unworthy.

134. HSPT-R10-022 - Answer A: Carbon dioxide
The second paragraph explicitly states that yeast uses sugars and releases carbon dioxide, which can occupy pockets inside the dough.

135. HSPT-R10-023 - Answer C: Gas production and a structure capable of retaining that gas both contribute.
The passage connects yeast activity with a stretchy, gas-retaining network and distinguishes these two requirements for increased volume.

136. HSPT-R10-024 - Answer D: Differences in gas retention can affect volume as well as yeast activity.
Volume reflects both gas production and retention; the same volume therefore does not uniquely determine the number of yeast cells.

137. HSPT-R10-025 - Answer C: It introduces the volume change, explains gas production and retention, and briefly follows the structure into baking.
The paragraphs move from the observed change to its internal processes, distinguish their roles, and then describe part of what baking changes.

138. HSPT-R10-026 - Answer A: Gas needs enclosing material that can stretch while still containing it.
The analogy represents gas pockets and their enclosing material; it clarifies the combined demands of expansion and containment without claiming literal rubber balloons.

139. HSPT-R10-027 - Answer D: Adhesion
The second paragraph directly labels the water–glass attraction adhesion and distinguishes it from water molecules attracting one another.

140. HSPT-R10-028 - Answer B: The water column’s weight balances the upward supporting effect.
The growing column becomes heavier until its weight balances the supporting effect at the curved surface, preventing unlimited upward rise.

141. HSPT-R10-029 - Answer B: To show that geometry changes how much water must be supported relative to the contact around the edge
The comparison ties the greater rise in a narrow tube to the relation between the supported water and the contact around its edge.

142. HSPT-R10-030 - Answer C: The same type of interaction can operate while different geometric arrangements impose different limits.
The final paragraph separates horizontal spreading from lifting a tall column, allowing the same general mechanism to have different limits.

143. HSPT-R10-031 - Answer A: routes or channels through which water can move
The surrounding discussion locates passages among fibers and describes water spreading through them, establishing the physical route meaning.

144. HSPT-R10-032 - Answer D: Surface interactions can support water, while weight and geometry limit the result.
The explanation combines supporting surface interactions with the growing column’s weight and the geometric differences among tubes and fiber spaces.

145. HSPT-R10-033 - Answer D: In the lower drawer beneath a yellow-striped cloth
The opening practical instructions locate spare rolls in the lower drawer under the cloth with the yellow stripe.

146. HSPT-R10-034 - Answer A: She has paid personal attention to the people who return to the carousel.
She knows the child’s explanation and the grandfather’s counting routine, revealing attention to individual visitors beyond operating the ticket punch.

147. HSPT-R10-035 - Answer D: small matters
The supposedly minor details concern visitors and a counterweight stone, rather than major operating rules, prizes, accusations, or contracts.

148. HSPT-R10-036 - Answer A: Her detailed memories and offer to return suggest attachment despite her denial.
The affectionate detail and volunteered return contradict the detached explanation she offers, making her denial transparently unconvincing.

149. HSPT-R10-037 - Answer C: It gradually reveals that handing over the job also means leaving familiar relationships and routines.
The shift expands a practical handover into a portrait of attachment, preparing for the final offer to return and its unpersuasive excuse.

150. HSPT-R10-038 - Answer D: One pass of the machine removes its visible outline.
The description says that the deep turning mark vanishes during resurfacing, before the first skater enters the renewed rink.

151. HSPT-R10-039 - Answer B: The quiet preparation of a used arena for the next period of activity
The narrative follows the surface's renewal and the first new skate mark, emphasizing the interval between past activity and what comes next.

152. HSPT-R10-040 - Answer B: The ice had never been used before that morning.
The narrative shows earlier marks being erased; their later absence therefore does not establish that no earlier skating occurred.

153. HSPT-R10-041 - Answer D: The next activity is beginning, while its particular course remains unwritten.
The curved mark becomes a figure for an unanswered future, extending the passage's movement from erased traces toward a new morning.

154. HSPT-R10-042 - Answer C: The sequence makes the renewed surface a transition between completed activity and an unknown next chapter.
The order connects visible remnants, their removal, and a new trace, giving the final image its sense of anticipation rather than mere damage.

155. HSPT-R10-043 - Answer B: The moving arm separates the contact surfaces.
The arm's movement both carries the striker toward the bell and separates the contact surfaces, interrupting current through the coil.

156. HSPT-R10-044 - Answer A: The moving arm repeatedly breaks and restores the coil's circuit.
The passage describes a cycle in which interrupted current releases the arm, the spring restores contact, and renewed current starts another pull.

157. HSPT-R10-045 - Answer A: The described break-and-return cycle would be prevented.
Contact separation supplies the interruption that removes magnetic pull; preventing it removes a necessary step in the particular repeating cycle described.

158. HSPT-R10-046 - Answer C: a conducting part that touches another part to complete a circuit
The following description identifies touching surfaces that complete the circuit, establishing contact as an electrical connecting part rather than a person or appointment.

159. HSPT-R10-047 - Answer B: To distinguish one continuing external request from the repeated internal switching it produces
The contrast explains how holding one control continuously can cause a sequence of separate electrical connections, releases, and mechanical strikes.

160. HSPT-R10-048 - Answer C: Every passenger in it has a priority ticket.
Vesa observes the gold stripe on everyone's ticket, and the clerk confirms that there are no ordinary passengers on this sailing.

161. HSPT-R10-049 - Answer B: She recognizes that a widely purchased upgrade might recreate the same lack of relative advantage.
The first upgrade provides no practical lead when everyone has it; her question tests whether the next upgrade could repeat that situation.

162. HSPT-R10-050 - Answer D: a special benefit
Vesa expects preferential boarding in return for the extra payment, making privilege a special benefit rather than a receipt, error, or schedule.

163. HSPT-R10-051 - Answer A: They show no physical progress despite the language of improved priority.
Vesa resumes her original place while the line remains still, contrasting unchanged movement with the promised improvement suggested by successive upgrades.

164. HSPT-R10-052 - Answer B: Vesa tests formal promises against their practical effect, while the clerk keeps restating categories and offering another category.
Each question exposes the gap between a category's stated privilege and its usefulness when all passengers share it, prompting another offered category.

165. HSPT-R10-053 - Answer C: They have remained in the same room long enough.
The opening specifies that the spoons have been left in the same room long enough, establishing the equal-temperature starting condition.

166. HSPT-R10-054 - Answer D: An object's initial temperature differs from the rate at which it transfers heat during touch.
The passage separates an initial state from heat-transfer behavior, explaining how equal starting temperatures can produce different sensations at the skin.

167. HSPT-R10-055 - Answer C: Make the faster-transferring object feel hotter rather than colder
When the objects start hotter than the skin, energy flows toward it; faster transfer can intensify the hot sensation, reversing the earlier cold comparison.

168. HSPT-R10-056 - Answer B: detects or responds to
The hand registers an interaction by responding to its thermal effect; the context concerns sensation rather than enrollment, payment, or identity.

169. HSPT-R10-057 - Answer D: To limit the explanation to heat-transfer direction rather than treating coldness as a permanent property of metal
The reversal shows that rapid transfer can intensify either cooling or warming, challenging the mistaken generalization that metal must always feel colder.

170. HSPT-R10-058 - Answer C: The doorways and the gaps between them become smaller farther back.
The opening describes smaller successive doorways and shrinking gaps as coordinated cues that exaggerate the impression of distance.

171. HSPT-R10-059 - Answer D: To explain how a viewpoint-dependent design can create theatrical depth and why it should be judged by its dramatic purpose
The passage combines an account of visual cues and viewing position with an argument about evaluating a stage set according to its intended function.

172. HSPT-R10-060 - Answer B: The spectator may see separations that make the set's physical construction more apparent.
Edges that align from one position can separate from another, so the changed view can expose how the illusion was arranged.

173. HSPT-R10-061 - Answer A: physically compressed in depth while suggesting greater depth from a selected view
The paragraph directly explains the shortened physical depth and the longer depth it is arranged to suggest from the intended position.

174. HSPT-R10-062 - Answer A: It presents an exaggerated literal belief that the audience need not hold to appreciate the illusion.
The deliberately excessive image distinguishes enjoying a convincing stage world from believing that its full physical counterpart exists behind the scenery.

Mathematics

175. HSPT-M10-001 - Answer C: −3 5/8
Moving right adds 3/8 to −4; writing −4 as −32/8 gives −29/8, or −3 5/8.

176. HSPT-M10-002 - Answer A: 13/48, 14/48, and 15/48 only
The endpoints are 12/48 and 16/48; only numerators 13, 14, and 15 lie strictly between them.

177. HSPT-M10-003 - Answer C: Two trailing zeros
A trailing zero needs a factor pair 2 × 5; the product contains two factors of 5 and enough factors of 2 to pair with both.

178. HSPT-M10-004 - Answer B: 11 and 12 only
The inequalities require n > 10 and n < 12.5; the only positive integers in that interval are 11 and 12.

179. HSPT-M10-005 - Answer D: The reading decreases by 8 units per minute.
The coefficient of t is −8, so each additional minute lowers C by 8; the initial value comes from the constant 90.

180. HSPT-M10-006 - Answer D: 8(x − 2) = 5(x + 1)
Multiplying both sides by 40 cancels each denominator, leaving 8 times the left numerator and 5 times the right numerator.

181. HSPT-M10-007 - Answer D: −12
Expanding gives ab + a + b + 1; substituting the known product and sum gives −10 − 3 + 1 = −12.

182. HSPT-M10-008 - Answer C: c = 6
The solutions are 4 + c and 4 − c; exactly one is negative when c > 4, which only 6 satisfies.

183. HSPT-M10-009 - Answer B: 9
Each triangular end has three edges, and three more join corresponding vertices of the ends, giving 3 + 3 + 3 = 9.

184. HSPT-M10-010 - Answer B: 20°
The exterior angles total 360°, so each ratio unit is 40°; the largest exterior is 160°, leaving the smallest interior, 20°.

185. HSPT-M10-011 - Answer A: 25π square units
The rectangle’s center is (0, 0); its distance to a vertex is $\sqrt{(3^2 + 4^2)} = 5$, giving circle area 25π.

186. HSPT-M10-012 - Answer A: It remains the same.
One centimeter equals 10 millimeters, so 12 centimeters and 120 millimeters specify exactly the same length in different units.

187. HSPT-M10-013 - Answer C: 14 inches
The print measures 320/80 = 4 inches by 240/80 = 3 inches; its perimeter is 2(4 + 3) = 14 inches.

188. HSPT-M10-014 - Answer A: The mean and median are 4, but there is no mode.
The mean is 12/3 = 4, and the middle value is 4; each value occurs once, so none is a mode.

189. HSPT-M10-015 - Answer D: 11/36
Keep a first-roll 6 with probability 1/6; otherwise obtain 6 on the second roll with probability (5/6)(1/6). Their sum is 11/36.

190. HSPT-M10-016 - Answer B: 1/5
There are nine equal-digit numbers and nine ending in zero, with no overlap; 18 favorable integers among 90 give probability 1/5.

191. HSPT-M10-017 - Answer D: 124
124 ÷ 4 = 31, but 124 ÷ 8 is not an integer; 128 and 136 are multiples of 8, while 130 is not divisible by 4.

192. HSPT-M10-018 - Answer D: 1.32
Thirteen tenths equal 1.3; adding two hundredths gives 1.3 + 0.02 = 1.32, with the excess tenths regrouped as a whole unit.

193. HSPT-M10-019 - Answer C: 63
Since $72 = 2^3 \times 3^2$, n must cancel both factors of 3. The largest multiple of 9 below 72 is 63, yielding 7/8.

194. HSPT-M10-020 - Answer B: 2a − b
Subtracting the parentheses changes both internal signs: a − b + a combines to 2a − b, not 2a + b.

195. HSPT-M10-021 - Answer A: Four: −5, −4, 4, and 5
The absolute value must be 4 or 5, and each permits either sign; strict inequality excludes both −3 and 3.

196. HSPT-M10-022 - Answer D: y = 60/x
Every column has product xy = 60; dividing by positive x gives y = 60/x, while the other equations fail later columns.

197. HSPT-M10-023 - Answer B: −2 < x ≤ 1 or x ≥ 4
The positivity conditions are x > −2, x > 1, and x < 4. Exactly two hold on (−2, 1] and [4, ∞).

198. HSPT-M10-024 - Answer C: Its vertex coordinates
A nonzero horizontal translation changes every vertex's x-coordinate, while preserving distances, angles, and the area enclosed by the triangle.

199. HSPT-M10-025 - Answer D: (−5, 0)
Relative to (1, 1), the point is (1, −6); clockwise rotation gives (−6, −1), and restoring the center gives (−5, 0).

200. HSPT-M10-026 - Answer B: 24 + 12√2 units
The original outer boundary contributes 24; the diagonal has length 6√2 and becomes an edge on both triangles, adding 12√2.

201. HSPT-M10-027 - Answer C: The 1.5-minute timer, by 25 seconds
The first timer runs 65 seconds and the second 90 seconds; 90 − 65 = 25 seconds, so the decimal-minute timer finishes later.

202. HSPT-M10-028 - Answer B: 0.3 millimeter
The stack is 36 millimeters thick; dividing by 120 cards gives 36/120 = 0.3 millimeter per card.

203. HSPT-M10-029 - Answer C: 30°C
A 100°Z increase corresponds to 200°C, so each °Z step is 2°C; 25°Z is −20 + 25 × 2 = 30°C.

204. HSPT-M10-030 - Answer A: 50%
Three of the six observations are strictly below 15: 10, 10, and 12. Thus the specified percentile rank is 3/6 = 50%.

205. HSPT-M10-031 - Answer A: $t \geq 8$
With $t \geq 8$, the ordered list's third value is 8; with $t < 8$, at least three observations are below 8.

206. HSPT-M10-032 - Answer A: 1/3: one of the three equally likely H-showing disk-face outcomes belongs to F.
Four disk-face outcomes are equally likely. Three show H, but only one of those belongs to F, giving conditional probability 1/3.

207. HSPT-M10-033 - Answer B: 100 times
The 5 on the left represents 5,000, while the 5 on the right represents 50; 5,000 ÷ 50 = 100.

208. HSPT-M10-034 - Answer A: −1/5
Distances from zero are absolute values: 0.2, 0.3, about 0.286, and 0.25; the smallest distance is 0.2, belonging to −1/5.

209. HSPT-M10-035 - Answer C: 19
The three position values are 16, 4, and 1; the numeral therefore means 1 × 16 + 0 × 4 + 3 × 1 = 19.

210. HSPT-M10-036 - Answer D: 30
A difference of square remainders cannot be 2 modulo 4, but 30 leaves 2; the others equal $6^2 - 1^2$, $7^2 - 1^2$, and $9^2 - 1^2$.

211. HSPT-M10-037 - Answer A: 3x = 9
Subtract corresponding sides: 5x − 2x = 3x, the equal 3y terms cancel, and 17 − 8 = 9.

212. HSPT-M10-038 - Answer D: x = 3
Since $27 = 3^3$, equal powers of 3 require $x = 3(x - 2)$; solving $x = 3x - 6$ gives $x = 3$.

213. HSPT-M10-039 - Answer C: 4
Substitution gives $(6x - 4) - 2(3x - 4) = 6x - 4 - 6x + 8 = 4$, independently of x.

214. HSPT-M10-040 - Answer B: There is no real solution.
Factoring reduces the equation to $x + 3 = 6$, whose only candidate is 3; that value is excluded by the original denominator.

215. HSPT-M10-041 - Answer A: The segment from A to B, including both endpoints
One ray continues beyond B and the other beyond A; only the segment connecting their starting points belongs to both rays.

216. HSPT-M10-042 - Answer B: 10 units
The opposite sums give 7 + 12 = 9 + x; subtracting 9 from 19 yields the missing side, 10 units.

217. HSPT-M10-043 - Answer A: π/6
The sphere has radius 3 and volume 36π; the cube's volume is 216, so the occupied fraction is 36π/216 = π/6.

218. HSPT-M10-044 - Answer D: 6 hours
The battery stores 12 × 5 = 60 watt-hours; dividing by a 10-watt rate gives 60/10 = 6 hours.

219. HSPT-M10-045 - Answer D: 3.6 liters
The flow is 12 × 5 = 60 cubic centimeters per second; multiplying by 60 and dividing by 1,000 gives 3.6 liters per minute.

220. HSPT-M10-046 - Answer B: The number of workers in each crew
Crew sizes turn the separate means into total minutes; adding those totals and dividing by the combined worker count determines the overall mean.

221. HSPT-M10-047 - Answer C: 12/24
Three nonadjacent position pairs can hold A and B, with two orders for each and two orders for C and D: 3 × 2 × 2 = 12 of 24.

222. HSPT-M10-048 - Answer C: 6 tags
Five tags can be one pair plus three singles. Six cannot avoid a second pair because each of four colors has only two tags.

223. HSPT-M10-049 - Answer C: $(-4)^2$
Only $(-4)^2$ places the negative sign inside the squared base, giving 16; the other placements retain the negative of 4^2.

224. HSPT-M10-050 - Answer A: 85 through 125
Common multiples are multiples of 42: 42, 84, 126, and so on. The interval strictly between 84 and 126 contains none.

225. HSPT-M10-051 - Answer D: No positive integer n can do this.
For 2n to be a square, n's exponent of 2 must be odd; for 3n to be a square, that same exponent must be even.

226. HSPT-M10-052 - Answer B: Multiply the 2 by 5 as well, giving 5x + 10.
The factor 5 applies to both terms inside the parentheses, so the constant contribution is 5 × 2 = 10.

227. HSPT-M10-053 - Answer C: (W − U − V)/2
Expanding W gives $x^2 + 2xy + y^2$; subtracting U and V leaves 2xy, which must be divided by 2.

228. HSPT-M10-054 - Answer A: (x + y)/(y − x)
The sum is (y + x)/(xy) and the difference is (y − x)/(xy); dividing cancels the common denominator and gives (x + y)/(y − x).

229. HSPT-M10-055 - Answer A: {−5, −1, 1, 5}
The inner expression |x| − 3 must equal 2 or −2, so |x| is 5 or 1; each allows both signs.

230. HSPT-M10-056 - Answer A: PR
The right angle is at Q; the hypotenuse is the opposite side, connecting the other two vertices, P and R.

231. HSPT-M10-057 - Answer D: 1 point
The reflected center is (−3, 0); the centers are 6 apart, equal to the two radii combined, so the circles touch only at the origin.

232. HSPT-M10-058 - Answer C: 30 cubic units
For edges a, b, c, the face areas are ab, bc, ac. Their product is $(abc)^2 = 6 \times 10 \times 15 = 900$, so volume abc = 30.

233. HSPT-M10-059 - Answer D: hours
Dividing kilometers by kilometers per hour cancels kilometers and leaves hours, consistent with the time required to cover the distance.

234. HSPT-M10-060 - Answer B: Its original direction
In four seconds the turntable completes 45 × 4/60 = 3 full revolutions; a whole number of turns returns the mark to its starting direction.

235. HSPT-M10-061 - Answer D: 48 square centimeters
Even the upper boundary dimensions, 8.5 and 5.5 centimeters, give only 46.75 square centimeters; an area of 48 is therefore impossible.

236. HSPT-M10-062 - Answer B: 8 seconds
Q1 = (3 + 5)/2 = 4 and Q3 = (11 + 13)/2 = 12; the middle-half spread is 12 − 4 = 8 seconds.

237. HSPT-M10-063 - Answer C: 1/12
Both lights are on only between minutes 3 and 4, a one-minute overlap within the 12-minute interval, giving probability 1/12.

238. HSPT-M10-064 - Answer B: 2/3; starting first gives A an advantage.
Let p be A's winning probability. A wins immediately with probability 1/2; two tails restart with probability 1/4. Thus p = 1/2 + p/4, giving p = 2/3.

Language

239. HSPT-L10-001 - Answer C: The shipment contains twenty-one brushes.
The compound number twenty-one takes a hyphen when written as words; a space, comma, or colon does not join it correctly.

240. HSPT-L10-002 - Answer C: D. R. Vega
D and R each replace a given name and therefore receive periods under the stated rule; no comma belongs between them and Vega.

241. HSPT-L10-003 - Answer D: "Two otters was [sic] resting."
Was is the original agreement error with plural otters; placing [sic] immediately after was preserves and identifies it without silently correcting it.

242. HSPT-L10-004 - Answer C: Keep the lid / closed during transport.
A line break within a sentence does not require a capital; these ordinary nouns and the continuing word closed remain lowercase.

243. HSPT-L10-005 - Answer D: Table 2 gives the dimensions of the table beside the door.
The numbered reference uses the specified capital T; the furniture description does not name a numbered report element and stays lowercase.

244. HSPT-L10-006 - Answer B: origami
Origami begins ori and ends gami; the alternatives change the first internal vowel or incorrectly alter the final syllables.

245. HSPT-L10-007 - Answer D: chassis
Chassis contains double s in the middle and ends is; the alternatives omit a middle s, add a final s, or change the vowel.

246. HSPT-L10-008 - Answer A: Turn them off.
With a pronoun object, this separable phrasal verb places the pronoun between turn and off; them is the required object form.

247. HSPT-L10-009 - Answer C: carry; to wait
Causative had followed by a person takes a bare infinitive; got followed by a person takes a to-infinitive.

248. HSPT-L10-010 - Answer D: to have been moved
The perfect passive infinitive to have been moved places the completed movement before the present belief and keeps the sculpture as the receiver.

249. HSPT-L10-011 - Answer A: Maybe; may be
Maybe is the adverb meaning perhaps; may be is the modal verb phrase needed after the subject passengers.

250. HSPT-L10-012 - Answer A: economic; economical
Economic relates to an economy or financial conditions; economical describes avoiding waste or expense, as the fuel-saving heater does.

251. HSPT-L10-013 - Answer B: However
However signals the limitation of the earlier benefit: correct sorting does not by itself make a file’s contents recognizable.

252. HSPT-L10-014 - Answer B: For instance, 2026-03-07-tide-notes identifies both the date and the subject.
The example retains the date pattern and adds tide-notes as a meaningful subject, directly illustrating the proposed improvement.

253. HSPT-L10-015 - Answer A: Immediately after sentence 4
The sentence explains why the two-digit requirement in sentence 4 matters, before sentence 5 illustrates that requirement.

254. HSPT-L10-016 - Answer B: The parts came from Rivet Inc.
The abbreviation’s period also closes this declarative sentence; a second period is unnecessary, and a comma or colon cannot replace the sentence ending.

255. HSPT-L10-017 - Answer C: "The engine is not ... ready for testing."
The correct omission removes the intervening comment but retains not; the alternatives remove the negation or rearrange words, changing the original claim.

256. HSPT-L10-018 - Answer A: She earned an A in algebra.
The letter grade is written A; ordinary algebra is not a proper course title or language name and stays lowercase.

257. HSPT-L10-019 - Answer D: At dusk, Patience found me without any patience.
The personified character’s proper name is Patience; the second occurrence names an ordinary quality rather than the character and stays lowercase.

258. HSPT-L10-020 - Answer D: safari
Safari uses one f and ends in i; the alternatives double the consonant or replace the final vowel with y.

259. HSPT-L10-021 - Answer C: sustenance
Sustenance has the sequence susten followed by ance; the alternatives change the internal vowels in this noun for nourishment or support.

260. HSPT-L10-022 - Answer A: fluorescent
Fluorescent begins fluor and ends escent; the alternatives reverse the first vowel sequence, change e to a, or transpose s and c.

261. HSPT-L10-023 - Answer C: might have
The contracted ending ’ve represents have; a modal such as might takes the base auxiliary have, not had, of, or is.

262. HSPT-L10-024 - Answer B: requires
Running heads a gerund phrase functioning as the subject; requires is the present-tense main verb that states what the activity demands.

263. HSPT-L10-025 - Answer A: distract
Lest introduces a feared consequence and can take the subjunctive base form; distract remains uninflected even though glare is singular.

264. HSPT-L10-026 - Answer B: bail; bale
Bail means remove water from a boat; bale names a tightly bound bundle, such as a bundle of hay.

265. HSPT-L10-027 - Answer D: carats; caret
Carats measure gemstone mass, whereas a caret is an editing mark showing an insertion point; the contexts require those separate terms.

266. HSPT-L10-028 - Answer B: Move sentence 3 to follow sentence 6.
Sentence 3 introduces a small surface detail before the overall arrangement is established; moving it later follows the promised general-to-detailed development.

267. HSPT-L10-029 - Answer A: The metal band
Sentence 6 contrasts the shiny element with the sphere’s dull surface; naming the metal band makes that reference precise rather than leaving an ambiguous pronoun.

268. HSPT-L10-030 - Answer D: A consistent viewpoint and an overall arrangement give listeners a framework for locating smaller features.
The revision brings together the shared viewpoint and broad arrangement, explaining how both help listeners place the smaller details.

269. HSPT-L10-031 - Answer B: Please tell me where the entrance is.
Please tell me makes the main clause a command; the embedded where clause does not turn the whole sentence into a direct question.

270. HSPT-L10-032 - Answer C: The faster the wheel turned, the louder the hum became.
The comma separates the two linked comparative clauses after turned; the alternatives interrupt a clause instead of marking the boundary between them.

271. HSPT-L10-033 - Answer D: The exhibit is closed. (It reopens on Monday.)
A separate parenthetical sentence starts with a capital and keeps its own period inside the closing parenthesis; the preceding sentence also ends independently.

272. HSPT-L10-034 - Answer A: Yours truly,
In this conventional closing, the first word is capitalized and truly is lowercase; the closing is not written in title case.

273. HSPT-L10-035 - Answer D: Only one thing mattered: The door must stay shut. They needed one thing: a key.
The door must stay shut is a complete sentence, while a key is a noun phrase; the stated style treats their initial letters differently.

274. HSPT-L10-036 - Answer B: trellis
Trellis has double l, a single final s, and the vowel sequence e then i; the alternatives alter one of those features.

275. HSPT-L10-037 - Answer C: semaphore
Semaphore contains ph, begins sema, and ends phore; the alternatives replace ph, change a to i, or transpose o and h.

276. HSPT-L10-038 - Answer D: mine
The construction a friend of mine uses the independent possessive pronoun mine; my requires a following noun, and mine takes no apostrophe.

277. HSPT-L10-039 - Answer B: The printer stopped printing.
Stopped printing ends the printing activity; stopped to print instead describes pausing another activity in order to begin printing.

278. HSPT-L10-040 - Answer C: It was such a heavy box that it took two people to lift it.
Before a singular count noun, this construction orders such + a + adjective + noun, giving such a heavy box.

279. HSPT-L10-041 - Answer A: sheer; shear
Sheer describes thin, partly transparent fabric; shear means cut the fleece from sheep, so the contexts require different homophones.

280. HSPT-L10-042 - Answer C: broach; brooch
Broach means introduce a topic for discussion; brooch names an ornamental pin worn on clothing, such as the silver accessory.

281. HSPT-L10-043 - Answer A: prosecute; persecute
Prosecute means pursue legal proceedings; persecute means subject someone to persistent unfair treatment, particularly because of identity or beliefs.

282. HSPT-L10-044 - Answer A: An Arrival the Audience Imagines
The title highlights the inferred destination created by alternating shots, rather than treating incidental bread, bicycle parts, or the letter as the central subject.

283. HSPT-L10-045 - Answer B: The final shot shows the cyclist delivering a letter on an entirely different street, thereby overturning the expectation created by the earlier editing.
Thereby explicitly links the different destination to the collapse of the earlier expectation, retaining both the event and its effect.

284. HSPT-L10-046 - Answer D: How bright the lantern is!
The how construction expresses an enthusiastic exclamation, so an exclamation point supplies the appropriate ending rather than a question mark, semicolon, or colon.

285. HSPT-L10-047 - Answer A: What the instrument measures is pressure.
What the instrument measures is the full subject of is; punctuation must not split that subject from its main verb or interrupt the subject clause.

286. HSPT-L10-048 - Answer B: "The bridge," Alban said, "will open at noon."
The starts the spoken sentence; will continues that same sentence after the inserted speech tag, so it does not receive a new initial capital.

287. HSPT-L10-049 - Answer C: A DIY Guide to Heat-Resistant Coatings
DIY retains its acronym capitals, internal to stays lowercase, and Heat-Resistant capitalizes both parts under the stated compound-title rule.

288. HSPT-L10-050 - Answer A: lariat
Lariat has one r, one final t, and the internal sequence ria; the alternatives transpose letters or unnecessarily double a consonant.

289. HSPT-L10-051 - Answer C: vellum
Vellum uses double l and the vowel u before one final m; the alternatives move the doubling or replace that vowel.

290. HSPT-L10-052 - Answer B: leisure
Leisure begins lei and ends sure; the alternatives reverse e and i, substitute z for s, or reverse the ending letters.

291. HSPT-L10-053 - Answer D: The volunteer will look after him.
Look after keeps its preposition before the object; him is the object form, so neither moving after nor using he is correct.

292. HSPT-L10-054 - Answer B: Almost all visitors signed the book.
Almost before all visitors limits the size of the signing group; placing it with signed instead limits completion of the action.

293. HSPT-L10-055 - Answer C: hangs
The postposed subject is the singular phrase a row, not plural lamps; the inverted word order therefore requires the singular verb hangs.

294. HSPT-L10-056 - Answer D: consul; console
Consul names an official representing a country abroad; console means comfort someone in distress, so the person and the action require different words.

295. HSPT-L10-057 - Answer C: grizzly; grisly
Grizzly names the kind of bear, while grisly describes something horrifying; their similar sounds do not make the spellings interchangeable.

296. HSPT-L10-058 - Answer A: disburse; disperse
Disburse means pay out money, whereas disperse means scatter or spread apart; funds and a departing crowd call for those distinct verbs.

297. HSPT-L10-059 - Answer A: During one design discussion, the doorway may need to be moved to several alternative positions.
Repeated doorway moves illustrate the need to undo and revise connections, directly supporting the value of reversibility during the working stage.

298. HSPT-L10-060 - Answer D: Once the layout is settled
Once the layout is settled marks the shift from revisable working model to display version, when concealed attachments can become a higher priority.